

Quantum Field Theory – 2
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Handwritten lecture notes, transcribed

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1 Lecture 1

Goals :—

- QFT 1 recap
 - Scalar & spinor fields
 - L & H formulation
 - canonical quantisation
 - symmetries & conservation laws
 - Interaction & S matrix
 - QED scattering examples
- Functional derivatives / Integrals
- Renormalisation
- Gauge theories (Gauge principle, parallel transport)
- Quantisation
 - Functional Integrals
 - BRST
 - Renormalisation.
- Spontaneous symmetry breaking, Anomalies
- Gauge theories (some non-perturbative questions)
 - Anomalies / Index theories
 - Confinement / Area law
 - t'Hooft & polyakov monopoles

Text books

- (1) Itzykson & Zuber
- (2) Weinberg V II
- (3) Srednicki QFT

(the three above) } Choose any one book.

- Peskin & Schroeder
- Schwartz
- Ramond.

Exams & Assignments –

- Assignments – every week or one per 10 days
 - Exams – Mid term, End term } 4 hrs or more if required.
-

Notations –

$$ds^2 = dt^2 - (d\vec{x})^2$$

Minkowski metric

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

$$p \cdot x = p_\mu x^\mu = p^\mu x_\mu = p_0 x_0 - \vec{p} \cdot \vec{x}$$

$$\epsilon^{ijk} = \epsilon^{123} = \underline{1} \quad \epsilon^{\mu\nu\rho\sigma} = \epsilon^{0123} = 1$$

∂ is sometimes used to denote boundary of space or space-time region. (So, ∂V or $\partial\Sigma$ are boundaries of V & Σ respectively.)

QFT–1 Recap

<u>Particle</u>	<u>Field</u>
(1) spin 0 (uncharged, massive)	scalar field
(2) spin 0 charged boson	complex scalar field
(3) photon (spin –1 massless)	$A_\mu(\vec{x}, t)$ real vector field / potential.
(4) spin $\frac{1}{2}$ fermions ($e^\pm, \gamma$, quarks ...)	spinor field $\Psi_r(\vec{x}, t)$

Scalars –

$$u_k(\vec{x}) = \frac{e^{-ik \cdot x}}{\sqrt{2\omega_k V}} \quad \omega_k = \sqrt{k^2 + m^2} \quad (V \rightarrow \infty \text{ in end})$$

$$\int d^3x \left(u_k^* \left(-i\partial_0 u_{k'} \right) - (i\partial_0 u_k^*) u_{k'} \right) = \delta_{k,k'}$$

where

$$k \equiv (k_1, k_2, k_3) = \frac{2\pi}{L} (n_1, n_2, n_3), \quad V = L^3$$

for $V \rightarrow \infty$, k becomes continuous &

$$\delta_{k,k'} \longrightarrow \frac{(2\pi)^3}{V} \delta(k - k')$$

$$\sum_k \longrightarrow V \int \frac{d^3k}{(2\pi)^3}$$

these wave fun. obey K.G. eqⁿ

$$(\partial^2 + m^2) u_k = 0 \quad \text{i.e.} \quad (\square + m^2) u_k = 0$$

K.G. eqⁿ comes from

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2$$

Fermions –

obey dirac eqⁿ –

$$(-i\gamma^\mu \partial_\mu + m) \Psi(x) = 0 \quad (\text{or}) \quad (i\not{\partial} - m) \psi = 0$$

where

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \mathbb{I}_{4 \times 4}$$

dirac eqⁿ comes from

$$\mathcal{L}_{\text{Dirac}} = \bar{\Psi} (i\not{\partial} - m) \Psi$$

Spinor field ‘ Ψ ’ can be expanded as

$$\Psi_r(x) = \sum_{p,r} \sqrt{\frac{m}{E_p V}} \left(a_{p,r} u_r(p) e^{-ip \cdot x} + b_{p,r}^\dagger v_r(p) e^{ip \cdot x} \right)$$

where u, v are solⁿ of dirac eqⁿ.

we will study about other fields later on.

Functional derivatives –

we know, Action

$$(S) = \int_{\Sigma} d^4x \mathcal{L}(\varphi, \partial_\mu \varphi)$$

$\Sigma \rightsquigarrow$ space time region.

Given a field configⁿ & its derivative we get a real number. So it is a functional.

S = Functional of field & its derivatives.

Note – Thinking of functions as expansion in sine & cosine fun. (Fourier series) is more important in QM & QFT than thinking it in terms of taylor series.

We expand ‘ φ ’ in terms of basis functions $\{f_n\}$

$$\varphi(x) = \sum_n c_n f_n(x)$$

specifying φ is same as specifying c_n .

$$(\varphi + \delta\varphi) \sim \sum (c_n + \delta c_n) f_n(x)$$

we are interested in changing φ in bulk / volumetric region of Σ & keeping φ fixed at surface / boundary represented by $\partial\Sigma$. i.e.

$$\partial\varphi|_{\partial\Sigma} = 0$$

$\partial\Sigma \rightsquigarrow$ represents boundary of region Σ .

$$\delta I = \int_{\Sigma} d^4x \sigma(x) \delta\varphi(x) \quad \left\{ \begin{array}{l} \sigma(x) \text{ is arbitrary fun.} \\ \text{depend upon } \varphi \text{ \& } \partial_\mu \varphi. \end{array} \right.$$

then functional derivative

$$\frac{\delta I}{\delta\varphi(x)} = \sigma(x) \quad (\text{defination})$$

ex :

$$I[\varphi, \partial_\mu \varphi] = \int_{\Sigma} d^4x \left(\frac{1}{2} (\partial_\mu \varphi)(\partial^\mu \varphi) - \frac{1}{2} m^2 \varphi^2 \right)$$

then

$$\frac{\delta I}{\delta \varphi(x)} = -(\partial^2 + m^2) \varphi(x)$$

then we require that,

$$\int_{\Sigma} d^4x \varphi^2(x) < \infty \quad \text{and} \quad \int_{\Sigma} d^4x (\partial \varphi)^2(x) < \infty$$

} To make action finite & finite energy $\int d^3x \mathcal{H} \rightarrow \text{finite}$.

$$\mathcal{F} = \{\text{all } \varphi\text{'s such that } \varphi : \Sigma \rightarrow \mathbb{R} \text{ with above cond}^n\}$$

A Functional (like action S/I) is a real valued function on $\mathcal{F}$ (where $\mathcal{F}$ is infinite dimensional).

Action S or I has physical significance.

if $S \sim \hbar$ use Q.M

if $\frac{S}{\hbar} \gg 1$ we can safely use C.M.

we also expand action as perturbation series of $\hbar$:

$$I = I_0 + \hbar I_1 + \hbar^2 I_2 + \dots$$

$I_0 \downarrow$ classical action; $I_1 \downarrow$ first order quantum correction $\dots$

Canonical quantisation –

(1) Scalar field

For

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \varphi)^2 - \frac{m^2 \varphi^2}{2}$$

we know

$$\begin{aligned} \Pi(\vec{x}, t) &= \partial_0 \varphi(\vec{x}, t) \\ H &= \frac{1}{2} \int d^3x (\Pi^2 + (\nabla \phi)^2 + m^2 \varphi^2) \end{aligned}$$

Imposing canonical commutation relations –

$$[\varphi(\vec{x}, t), \Pi(\vec{x}', t)] = i \delta^3(\vec{x} - \vec{x}')$$

all other equal time commutator vanishes.

(2) Dirac Field –

$$\begin{aligned}\mathcal{L}_0 &= \bar{\Psi} (i\not{\partial} - m) \Psi \\ \Pi &= i\psi^\dagger (\gamma^0)^2 = i\psi^\dagger \quad \dots \quad \{(\gamma^0)^2 = +\mathbb{I}\} \\ H &= \int d^3x \psi^\dagger (i\gamma^0 \gamma^i \partial_i + m\gamma^0) \psi\end{aligned}$$

H is hermitian (we expect it to be).

$$H = \sum_{p,r} E_p \left(a_{p,r}^\dagger a_{p,r} + b_{p,r}^\dagger b_{p,r} \right) - \boxed{2 \sum_p E_p}$$

✓ Becomes infinity. Just throw it.

We can't impose canonical commutation relations, as H is unbounded from below. So we use anti-commutation relations. {also $\Psi(x)$ should be valued in Grassmann no.s}.

Impose anti-commutation relations –

$$\{\Psi_r(\vec{x}, t), \Pi_s(\vec{x}', t)\} = i \delta_{r,s} \delta(\vec{x} - \vec{x}')$$

(Others are zero.)

(No class – Next week)

2 Lecture 2

Propagators –

(1) Free fields

$$\begin{aligned}\varphi(x) &= \int \frac{d^3k}{(2\pi)^3} \left(a_k e^{-ik \cdot x} + a_k^\dagger e^{ik \cdot x} \right) \\ [\varphi(x), \varphi(y)] &= \int d\mu_k \left(e^{-ik \cdot (x-y)} - e^{ik \cdot (x-y)} \right)\end{aligned}$$

↓ a fun. rather than operator

$$\begin{aligned}\int \frac{d^4k}{(2\pi)^4} \frac{-i}{(k^2 - m^2)} e^{-ik \cdot (x-y)} \\ \equiv \Delta(x-y)\end{aligned}$$

since $[\varphi(x), \varphi(y)]$ obeys (Translational Invariance).

Green's fun.

$G(x, y)$ obeys

$$\begin{aligned}(\square_x + m^2) G(x, y) &= -i \delta^4(x - y) \\ G(x, y) &= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2} e^{-ik \cdot (x-y)}\end{aligned}$$

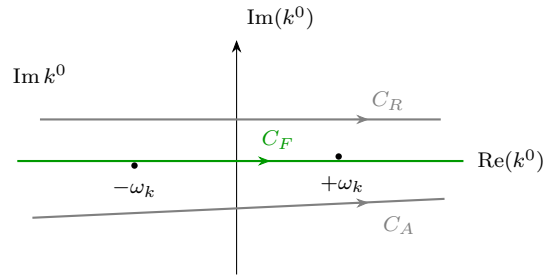
$$\begin{aligned}
& \rightarrow (k^0)^2 - (\vec{k}^2 + m^2) \\
& = \int \frac{dk^0}{2\pi} \int \frac{d^3k}{(2\pi)^3} \frac{i}{(k^0)^2 - (\vec{k}^2 + m^2)} e^{-i(k \cdot (x-y))}
\end{aligned}$$

Choice of contour for k^0 integral is important.

$G_R(x, y)$ = retarded greens fun.

$G_A(x, y)$ = advanced greens fun.

$G_F(x, y)$ = Feynman's Green's fun. (Feynman propagator).



$$G_F(x, y) = \langle 0 | T(\varphi(x)\varphi(y)) | 0 \rangle$$

More generally we are interested in –

$$G(x_1, x_2, x_3 \dots x_n) = \langle 0 | T(\varphi(x_1)\varphi(x_2) \dots \varphi(x_n)) | 0 \rangle$$

we can write generating functional

$$\begin{aligned}
Z[J] &= \sum_N \frac{1}{N!} \int d^4x_1 d^4x_2 \dots d^4x_N G(x_1, x_2 \dots x_N) J(x_1)J(x_2) \dots J(x_N) \\
&= \langle 0 | T e^{\int J(x)\varphi(x)} | 0 \rangle
\end{aligned}$$

J = Source function.

Z obeys the eqⁿ

$$(\partial_x^2 + m^2) \frac{\delta Z[J]}{\delta J(x)} = \langle 0 | T \left[(\partial_x^2 + m^2) \varphi(x) e^{\int J\varphi} \right] | 0 \rangle = -i J(x) Z[J]$$

Free fields –

$$(\partial_x^2 + m^2) \frac{\delta Z_0[J]}{\delta J(x)} = -i J(x) Z_0[J]$$

$$Z_0[J] = \mathcal{N} e^{\frac{1}{2} \int d^4x d^4y J(x) G(x,y) J(y)}$$

Derivation (marginal working)

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2} - \lambda\phi^4 \quad (1)$$

$$= \underbrace{-\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2}}_{\mathcal{L}_0} - J\phi - \lambda\phi^4 \Big|_{J=0}$$

$$\langle 0|0\rangle_J = \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi - \lambda\phi^4)$$

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}(\partial\phi)^2 - \frac{m^2\phi^2}{2} - J\phi \\ &= \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi) \underbrace{\exp i \int d^4x (-\lambda\phi^4)}_{\text{we will expand}} \\ &= \sum_n \frac{(-i\lambda)^n}{n!} \int D\phi \left(\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right) \int (\phi^4)^n d^4x \end{aligned} \quad (2)$$

let's see.

$$\begin{aligned} \frac{\delta}{\delta J(y)} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] &= i \int d^4x \frac{\delta}{\delta J(y)} [\mathcal{L}_0 - J(x)\phi(x)] \times \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= \left[i \int d^4x -\delta(x-y) \phi(x) \right] \times \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= -i \phi(y) \exp i \int d^4x (\mathcal{L}_0 - J\phi) \end{aligned} \quad (3)$$

On Generalizing.

$$\frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] = (-i)^{4n} \phi^{4n}(y) \exp i \int d^4x [\mathcal{L}_0 - J\phi]$$

we got

$$\phi^{4n}(y) \exp i \int d^4x (\mathcal{L}_0 - J\phi) = \frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] \quad (4)$$

use (4) in (2)

$$\begin{aligned} &= \sum_n \frac{(-i\lambda)^n}{n!} \int D\phi \int \frac{\delta^{4n}}{\delta J(y)^{4n}} \left[\exp i \int d^4x (\mathcal{L}_0 - J\phi) \right] \\ &= \sum_n \frac{(-i\lambda)^n}{n!} \int \left(\frac{\delta}{\delta J(y)} \right)^{4n} \int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi) \\ &= \sum_n \frac{\left[\int -i\lambda \left(\frac{\delta}{\delta J(y)} \right)^4 \right]^n}{n!} \underbrace{\int D\phi \exp i \int d^4x (\mathcal{L}_0 - J\phi)}_{Z_0(J)} \\ &= \left[\exp \int \left(-i\lambda \frac{\delta^4}{\delta J(y)^4} \right) \right] Z_0(J) \end{aligned}$$

Interactions –

$$\mathcal{L} = \frac{1}{2}(\partial\varphi)^2 - \frac{1}{2}m^2\varphi^2 - \lambda\varphi^4$$

E.O.M :

$$(\partial_x^2 + m^2) \varphi(x) + 4\lambda\varphi^3(x) = 0$$

$$\begin{aligned} Z[J] &= \mathcal{N} e^{-i\lambda \int d^4x \left(\frac{\delta}{\delta J(x)}\right)^4} Z_0[J] \\ &= e^{\frac{1}{2} \int J(x)G(x,y)J(y)} e^{\int J(x)G(x,y)\frac{\delta}{\delta\varphi(y)}} e^{\frac{1}{2} \int G(x,y)\frac{\delta}{\delta\varphi(x)}\frac{\delta}{\delta\varphi(y)}} e^{i\lambda \int \varphi^4(x) d^4x} \quad (?) \end{aligned}$$

where

$$\begin{aligned} V &= \frac{\delta}{\delta\varphi(z_1)} \cdots \frac{\delta}{\delta\varphi(z_n)} \mathcal{F}[\varphi] \quad (?) \\ \mathcal{F}[\varphi] &= \mathcal{N} \exp\left(\frac{1}{2} \int G \frac{\delta}{\delta\varphi(x)} \frac{\delta}{\delta\varphi(y)}\right) \exp(S_{\text{int}}) \quad \left(S_{\text{int}} = \int \lambda\varphi^4\right) \end{aligned}$$

LSZ Formulation –

$$V(x_1 x_2 \cdots x_n) = \prod_j \int i \left(\partial_{x_j}^2 + m^2 \right) G(x_1, x_2 \cdots x_n)$$

Scattering amplitude (S) :

$$\begin{aligned} S(k_1 \cdots k_n \longrightarrow k_{n+1} \cdots k_N) \\ = \prod_{j=1}^n \int u_{k_j}(x_j) i \left(\partial_{x_j}^2 + m^2 \right) \prod_{r=n+1}^N u_{k_r}^*(x_r) i \left(\partial_{x_r}^2 + m^2 \right) G(x_1 x_2 \cdots x_N) \end{aligned}$$

where $x_j = (x_j^0, \vec{x}_j)$.

Electromagnetic field –

$$\begin{aligned} S &= -\frac{1}{4} \int d^4x (F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu) \\ &= \frac{1}{2} \int d^4x \left[(\vec{E}^2 - \vec{B}^2) - A_0 J_0 + \vec{A} \cdot \vec{J} \right] \end{aligned}$$

where,

$$E_i = F_{0i}, \quad B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}$$

Maxwell's eqⁿ –

$$\begin{aligned} \partial_\mu F^{\mu\nu} &= J^\nu \\ \partial_\mu \left(\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \right) &= 0 \end{aligned}$$

A_μ has a gauge symmetry, i.e. $A_\mu \mapsto A_\mu + \partial_\mu \theta(t, x)$ gives same $F_{\mu\nu}$, i.e. ($\vec{E}$ & $\vec{B}$).

redundant eqⁿ –

$$\partial_i E_i = J^0 \quad (\text{Not a time evolution eq}^n).$$

Fix the gauge (or getting rid of redundancies) : one choice is to choose $A^0 = 0$. (A is a 4 vector in other frame A^0 can take non zero value, so there is some tension b/w lorentz invariance & fixing the gauge.)

then split,

$$A_i \equiv A_i^T + A_i^L$$

$A^L \longrightarrow \partial_i f$, and A^T obeys

$$\partial_i A_i^T = 0 \quad \xrightarrow{\text{in } \vec{k} \text{ space}} \quad (\vec{k} \cdot \vec{A}^T = 0)$$

$$E_i = \partial_0 A_i - \partial_i A_0^0 = \partial_0 A_i^T + \partial_i (\partial_0 f)$$

$$\Rightarrow \partial_i E_i = \partial_i \partial_i (\partial_0 f) = J_0$$

$$(\partial_0 f)(x^0, \vec{x}) = \int d^3 y \left[G_c(\vec{x} - \vec{y}) \right] J_0(x^0, \vec{y})$$

$$\downarrow \quad -\frac{1}{4\pi} \frac{1}{|\vec{x} - \vec{y}|}$$

$$S = \frac{1}{2} \int d^4 x [(\partial_0 A_j^T)(\partial_0 A_j^T) - (\partial_i A_j^T)(\partial_i A_j^T)] \\ + \int d^4 x \int d^4 y \frac{1}{2} J_0(x) G_c(x, y) J_0(y) \delta(x^0 - y^0) + \int d^4 x A_i^T J_i$$

The condition $A^0 = 0 = \vec{\nabla} \cdot \vec{A}$ is called radiation gauge. This is action for 2 massless field (the two transverse poles of A_i).

If J^μ is zero, then we have

$$\partial^2 A_i^T = 0$$

plane wave solⁿ e^{-ikx}

it has solutions

$$A_i^T(x) = \sum_{k, \lambda} \left(a_{k\lambda} e_i^{(\lambda)}(k) u_k(x) + c.c \right)$$

$\downarrow$ polarisation vectors

$$\sum_{\lambda=1,2} e_i^{(\lambda)}(k) e_j^{(\lambda)}(k) = \left(\delta_{ij} - \frac{k_i k_j}{\vec{k}^2} \right)$$

There wouldn't be much necessary for the things to come!

3 Lecture 3

Quantisation (QED) :

$$[a_{k\lambda}, a_{k'\lambda'}^\dagger] = i \delta_{kk'} \delta_{\lambda\lambda'} \quad (\hbar = 1)$$

Propagator :

$$\mathcal{D}_{ij}(x, y) = \langle 0 | T (A_i^T(x) A_j^T(y)) | 0 \rangle \\ \mathcal{D}_{ij}(x, y) = \int \frac{d^4 k}{(2\pi)^4} \left(\delta_{ij} - \frac{k_i k_j}{\vec{k}^2} \right) \frac{e^{-ik(x-y)}}{k^2 + i\varepsilon}$$

covariant form

$$= \eta_{\mu\nu} \int \frac{d^4 k}{(2\pi)^4} \frac{e^{-ik(x-y)}}{k^2 + i\varepsilon}$$

The S-matrix Functional. :

$$\mathcal{F} = \left(e^{-\frac{1}{2} \int_x \int_y \mathcal{D}_{\mu\nu}(x,y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)}} \right) e^{i A_\mu J^\mu}$$

$$\{ \mathcal{F}|_{A=0} = Z_0(J) \}$$

Scalar QED

$$\mathcal{L} = (\mathcal{D}_\mu \varphi) (\mathcal{D}^\mu \varphi)^* - m^2 \varphi \varphi^* + \mathcal{L}_{EM}$$

$$\mathcal{D}_\mu = \partial_\mu - iQ A_\mu$$

$$\mathcal{L}_{\text{int}} = -iQ \underbrace{A^\mu (\varphi^* \partial_\mu \varphi - \varphi \partial_\mu \varphi^*)}_{J^\mu} + Q^2 A^\mu A_\mu \varphi^* \varphi$$

The S-matrix Functional:

$$\mathcal{F} = \underbrace{e^{\left(\iint G(x_1, x_2) \frac{\delta}{\delta \varphi(x_1)} \frac{\delta}{\delta \varphi^*(x_2)} \right)}}_{(1)} e^{\left(-\frac{1}{2} \iint \mathcal{D}_{\mu\nu}(x, y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)} \right)} e^{i S_{\text{int}}}$$

$$S_{\text{int}} = \int d^4x \mathcal{L}_{\text{int}}$$

(1) & $\mathcal{L}_{\text{int}}$ is not gauge invariant but whole expression is gauge invariant.

Spinor QED –

$$\mathcal{L} = \bar{\Psi} (i \not{\partial} - m) \Psi + \mathcal{L}_{EM}$$

$$= \bar{\Psi} (i \not{\partial} - m) \Psi + e \bar{\Psi} \gamma^\mu A_\mu \Psi + \mathcal{L}_{EM}$$

$$\mathcal{F} = \exp \left(-\frac{1}{2} \iint \mathcal{D}_{\mu\nu}(x, y) \frac{\delta}{\delta A_\mu(x)} \frac{\delta}{\delta A_\nu(y)} \right) \cdot \exp \left(- \iint \frac{\delta}{\delta \bar{\Psi}_r(x)} S_{r,s}(x, y) \frac{\delta}{\delta \Psi_s(y)} \right) e^{ie \int A_\mu \bar{\Psi} \gamma^\mu \Psi}$$

Functional integral representation (of $\mathcal{F}$) :

Basis of φ {set of all f^n 's in which we expand φ }

Consider $\varphi(x)$ and $\varphi(x) + \delta\varphi(x)$

$$ds^2 \stackrel{\text{defn}}{=} \|\delta\varphi\|^2 = \int_\Sigma d^4x (\delta\varphi)^2$$

S : action

$$= \sum_n (\delta c_n)^2$$

say ds^2 is metric, where metric depends upon φ .

say $\varphi : \Sigma \longrightarrow S^2$ ($\Sigma \downarrow$ space-time region; $S^2 \rightsquigarrow$ sphere)

if $(\theta, \alpha) \in S^2$ then, φ can be thought of — functions $\theta(x)$ & $\alpha(x)$, ($x \in \Sigma$)

$$\varphi + \delta\varphi \sim (\theta + \delta\theta(x), \alpha(x) + \delta\alpha(x))$$

$$ds^2(\varphi, \varphi + \delta\varphi) = \int_{\Sigma} d^4x \left[(\delta\theta)^2 + (\sin^2 \theta) (\delta\alpha)^2 \right]$$

The set $\{c_n\} \equiv$ local coordinates on $\mathcal{F}$ (set of basis functions).

First consider only n -modes :

$$dV^{(N)} = \sqrt{|\det(g^{(N)})|} dc_1 dc_2 \cdots dc_N$$

$\downarrow N \times N$ matrix

$$I = \int [d\varphi] e^{-\frac{1}{2} \int_x \int_y \varphi(x) M(x,y) \varphi(y)} \quad \text{where} \quad \varphi(x) = \sum c_n f_n(x)$$

$$\text{exponent} = \sum_{mn} c_m M_{mn} c_n$$

$$M_{mn} = \int_{\Sigma} \int_{\Sigma} d^4x d^4y f_m(x) M(x,y) f_n(y)$$

We need M_{mn} to be diagonalizable, $\text{Re}(\text{eigen values}) > 0$.

Consider a function $u_n(x)$ defined by

$$\int d^4y M(x,y) u_n(y) = \lambda_n u_n(x)$$

then write $\varphi \sim \sum a_n u_n$

then,

$$\|\delta\varphi\|^2 = \sum_n (\delta a_n)^2 \quad dV = \prod_n^N da_n$$

$$\begin{aligned} I &= \lim_{N \rightarrow \infty} \int \cdots \int \prod_n^N da_n e^{-\frac{1}{2} \sum \lambda_n a_n^2} \quad (\text{Re}(\lambda) > 0) \\ &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(\frac{2\pi}{\lambda_n} \right)^{1/2} \equiv \left(\det \frac{M}{2\pi} \right)^{-1/2} \end{aligned}$$

Truncating to finite N , then taking $N \rightarrow \infty$ is called a Normalisation procedure.

Similarly consider,

$$I[J] = \int [d\varphi] e^{-\frac{1}{2} \int_{\Sigma} \int_{\Sigma} \varphi(x) M(x,y) \varphi(y) + \int_{\Sigma} J(x) \varphi(x)}$$

Completing the square

$$\begin{aligned} I &= \int [d\varphi] e^{-\frac{1}{2} \iint (\varphi - JM^{-1}) M (\varphi - M^{-1}J)} e^{\frac{1}{2} \int JM^{-1}J} \\ &= \left| \frac{M}{2\pi} \right|^{1/2} e^{\frac{1}{2} \iint J(x) M^{-1}(x,y) J(y)} \end{aligned}$$

If φ is complex, then

$$I[J, \bar{J}] = \left(\det \frac{M}{2\pi} \right)^{-1} e^{\iint \bar{J} M^{-1} J}$$

Functional integral repⁿ for fermions –

Grassmann variables (or numbers) are anti-commuting object on which complex conjugate like operation is also defined.

If α, β are grassmann numbers then $\alpha\beta$ & $\beta\alpha$ are also grassmann numbers.

$$\alpha\beta = -\beta\alpha \quad \Rightarrow \quad \{\alpha, \beta\} = 0$$

$$\alpha\beta + \beta\alpha = 0 \quad (\text{Sum of two grassman numbers is zero.})$$

For single grassmann variable η

$$\eta^2 = 0 \quad \text{i.e.} \quad \eta\eta + \eta\eta = 0 \Rightarrow 2\eta^2 = 0 \Rightarrow \eta^2 = 0$$

functions may be expanded as

$$f(\eta) = f_0 \eta^0 + f_1 \eta + \cancel{f_2 \eta^2} + \dots$$

if a function depends upon ordinary variable x & grassmann variable η .

$$f(x, \eta) = f_0(x) \eta^0 + f_1(x) \eta + \cancel{f_2(x) \eta^2} + \dots$$

If a fⁿ depends upon two grassmann numbers & an ordinary variable. (η_1, η_2) .

$$\eta_1^2 = 0 = \eta_2^2 ; \quad (\eta_1 \eta_2 = -\eta_2 \eta_1)$$

$$f(x, \eta_1, \eta_2) = f_0(x) + f_1(x) \eta_1 + f_2(x) \eta_2 + f_{12}(x) \eta_1 \eta_2 + 0$$

$$\bar{\eta} A \eta \quad \left\{ \bar{\eta} = (\bar{\eta}_1, \bar{\eta}_2), \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \right\}$$

$A \longrightarrow 2 \times 2$ matrix with $\mathbb{R}$ (or $\mathbb{C}$) entries

Lattice discretisation (working pages)

[These pages continue in the notebook after a blank leaf; they form one rough working block on discretising the Klein–Gordon operator on a lattice and are transcribed in the order written.]

$$\begin{aligned} \delta^2 \phi &= \frac{1}{a} \left(\frac{\phi(n + \hat{\mu}) - \phi(n)}{a} - \frac{(\phi(n) - \phi(n - \hat{\mu}))}{a} \right) \\ &= \frac{1}{a^2} (\phi(n + \hat{\mu}) - \phi(n) - \phi(n) + \phi(n - \hat{\mu})) \\ \delta^2 \phi &= \frac{1}{a^2} (\phi(n + \hat{\mu}) + \phi(n - \hat{\mu}) - 2\phi(n)) \end{aligned}$$

with $\phi = e^{ik \cdot x}$, $\phi(n) = e^{ik \cdot n}$:

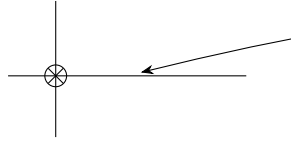
$$\begin{aligned} &= e^{ip \cdot (n + \hat{\mu})} + e^{ip \cdot (n - \hat{\mu})} - 2e^{ip \cdot n} \\ &= e^{ip \cdot n} (e^{ip \cdot \hat{\mu}} + e^{-ip \cdot \hat{\mu}} - 2) \\ &= e^{ip \cdot n} (e^{ip_\mu} + e^{-ip_\mu} - 2) \end{aligned}$$

$$\begin{aligned}
&= e^{ip \cdot n} (2 \cos p_\mu - 2) \\
&= -2e^{ip \cdot n} (1 - \cos p_\mu) \\
&= \left(4e^{ip \cdot n} \sin^2 \frac{p_\mu}{2} + m^2 \right)^{-1} \\
S &= \int \bar{\Psi} M \Psi d^4x
\end{aligned}$$

(margin, in blue:)

$$(\square_x + m^2) G(x, y) = \delta(x - y) ; \quad (\partial_x^2 + m^2) G(x, y) = \delta(x - y)$$

at $p_\mu = 0$,



$$\begin{aligned}
&\frac{1}{\sum_{\mu=1}^4 \sin^2 \frac{p_\mu}{2} + m^2} \\
&\frac{2\pi}{L} (n_1, n_2, n_3) \qquad \frac{2\pi}{L} n_1
\end{aligned}$$

H.W : No doubling in case of scalars, but we see doubling in case of fermions for massless case.

$$(\partial^2 + m^2) \phi(x) \qquad \phi(x) = \langle x | \phi \rangle$$

insert $\mathbb{I}$:

$$\begin{aligned}
&\langle x | \int dp \, |p\rangle \langle p| \phi \rangle \\
&(\partial^2 + m^2) \int dp \, \langle x | p \rangle \hat{\phi}(p) \\
\delta^2 &= \frac{1}{a^2} (\phi(n + \hat{\mu}) + \phi(n - \hat{\mu}) - 2\phi(n))
\end{aligned}$$

$$G(x,y) \,=\, G(|x-y|) \,=\, \int \frac{d^4k}{(2\pi)^4} \, e^{ik\cdot(x-y)} \, \widetilde{G}(k)$$

$\langle k|x-y\rangle$

$$\begin{aligned}
(\square_x + m^2) G(x, y) &= \int (-k^2 + m^2) \frac{d^4k}{(2\pi)^4} e^{ik \cdot (x-y)} \widehat{G}(k) \\
&= \delta^{(4)}(x - y) = \int \frac{d^4k}{(2\pi)^4} e^{ik \cdot (x-y)} \\
(-k^2 + m^2) \widehat{G}(k) &= 1 \quad \Rightarrow \quad \widehat{G}(k) = \frac{1}{-k^2 + m^2}
\end{aligned}$$

$$e^{ik \cdot (x-y)} \qquad \text{with} \quad x = na, \quad y = ma$$

$$\longrightarrow e^{ip \cdot (n-m)}$$

$$\phi(x) = \langle x | \phi \rangle = \int \frac{d^4 p}{(2\pi)^4} \hat{\phi}(p) e^{ip \cdot x}$$

$$\phi(n) = \int \frac{dp}{(2\pi)^4} \hat{\phi}(p) e^{ip \cdot n}$$

$$\phi(n + \hat{\mu}) = \int dp \hat{\phi}(p) \boxed{e^{ip \cdot (n + \hat{\mu})}}$$

$$S[\phi, \partial_\mu \phi] = \int d^4 x \phi \left(\int \frac{d^4 p}{(2\pi)^4} \hat{\phi}(p) \left[e^{ip \cdot (n + \hat{\mu})} + e^{ip \cdot (n - \hat{\mu})} - 2e^{ip \cdot n} \right] \right)$$

$\underbrace{\hspace{1cm}} \downarrow$ Inverse of propagator

$$\mathcal{D}_F(x - y) = \int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik \cdot (x - y)}}{k^2 - m^2}$$

4 Lecture 4

Assignment – 1

- (1) Read chapter 6, 7, 8 — for path integrals
43, 44 — path integral for fermions
- (2) Solve all problems in above chapters.

Functional integrals for fermions :

$$\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \quad a_{ij} \in \mathbb{R} \text{ or } \mathbb{C}$$

$$\bar{\eta} A \eta = (\bar{\eta}_1 \quad \bar{\eta}_2) \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$= (\bar{\eta}_1 \quad \bar{\eta}_2) \begin{pmatrix} a_{11}\eta_1 + a_{12}\eta_2 \\ a_{21}\eta_1 + a_{22}\eta_2 \end{pmatrix}$$

$$\bar{\eta} A \eta = \bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2$$

$$e^{-\bar{\eta} A \eta} = 1 - \bar{\eta} A \eta + \frac{(\bar{\eta} A \eta)^2}{2!} - \frac{(\bar{\eta} A \eta)^3}{3!} + \dots$$

$$= 1 - \underbrace{(\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2)}_{\textcircled{1}}$$

$$\textcircled{2} \left\{ \begin{array}{l} + (\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2) \\ \times (\bar{\eta}_1 a_{11} \eta_1 + \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{21} \eta_1 + \bar{\eta}_2 a_{22} \eta_2) \\ + \dots \end{array} \right.$$

$\hookrightarrow$ 2nd term contains $\bar{\eta}_2^2, \bar{\eta}_1^2, \eta_1^2\eta_2^2$ which all vanishes.

② = The terms which survive are

$$\begin{aligned}
& \bar{\eta}_1 a_{11} \eta_1 \bar{\eta}_2 a_{22} \eta_2 + \bar{\eta}_1 a_{12} \eta_2 \bar{\eta}_2 a_{21} \eta_1 \\
& + \bar{\eta}_2 a_{21} \eta_1 \bar{\eta}_1 a_{12} \eta_2 + \bar{\eta}_2 a_{22} \eta_2 \bar{\eta}_1 a_{11} \eta_1 \\
= & a_{11} a_{22} \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 + a_{12} a_{21} \bar{\eta}_1 \eta_2 \bar{\eta}_2 \eta_1 + a_{21} a_{12} \bar{\eta}_2 \eta_1 \bar{\eta}_1 \eta_2 + a_{22} a_{11} \bar{\eta}_2 \eta_2 \bar{\eta}_1 \eta_1 \\
= & a_{11} a_{22} (\bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 + \bar{\eta}_2 \eta_2 \bar{\eta}_1 \eta_1) + a_{12} a_{21} (\bar{\eta}_1 \eta_2 \bar{\eta}_2 \eta_1 + \bar{\eta}_2 \eta_1 \bar{\eta}_1 \eta_2) \\
= & \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 (a_{11} a_{22} + a_{11} a_{22} + a_{12} a_{21} (-1) + a_{12} a_{21} (-1)) \\
= & \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2 \cdot 2 \underbrace{(a_{11} a_{22} - a_{12} a_{21})}_{\det A}
\end{aligned}$$

So,

$$e^{-\bar{\eta} A \eta} = 1 - \bar{\eta} A \eta + (\det A) \bar{\eta}_1 \eta_1 \bar{\eta}_2 \eta_2$$

$$\underline{Z} = 0 \rightsquigarrow \psi(x) \begin{cases} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{cases} \longrightarrow \text{Grassmann co.}$$

$$\psi_1^2 = 0, \quad \underline{\psi_1 \psi_2 = -\psi_2 \psi_1}, \quad \psi_2^2 = 0$$

Recall Z for SHO

$$\begin{aligned}
Z &= \text{Tr} e^{-\beta H} = \sum_{n=0}^{\infty} \langle n | e^{-\beta H} | n \rangle \\
&= \sum_{n=0}^{\infty} e^{-\beta n (1 + \frac{1}{2}) \hbar \omega} = \frac{1}{2 \sinh \frac{\hbar \omega}{2}}
\end{aligned}$$

If we use position basis instead of energy eigenvector basis.

$$\begin{aligned}
Z &= \int dx_0 \langle x_0 | e^{-\beta H} | x_0 \rangle \quad \text{where, } \epsilon = \frac{\beta}{N} \\
&= \int dx_0 \langle x_0 | e^{-\epsilon H} \underbrace{\quad}_I e^{-\epsilon H} \underbrace{\quad}_I \cdots e^{-\epsilon H} | x_0 \rangle
\end{aligned}$$

Inserting $N - 1$ complete set of $|x\rangle$ states

$$\begin{aligned}
I &= \int dx |x\rangle \langle x| \\
&= \int dx_0 \langle x_0 | e^{-\epsilon H} \left| \int dx_1 |x_1\rangle \langle x_1| e^{-\epsilon H} \int dx_2 |x_2\rangle \langle x_2| \cdots \int dx_{N-1} |x_{N-1}\rangle \langle x_{N-1}| x_0 \right\rangle \\
Z &= \int dx_0 dx_1 dx_2 \cdots dx_{N-1} \prod_{i=0}^{N-1} \langle x_i | e^{-\epsilon H} | x_{i+1} \rangle \quad \text{with } (x_N = x_0)
\end{aligned}$$

let us find $\langle x|e^{-\epsilon H}|y\rangle$

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$e^{-\left(\epsilon \frac{p^2}{2m} + \epsilon V(x)\right)} = e^{-\epsilon \frac{p^2}{2m}} e^{-\epsilon V(\hat{x})} + O(\epsilon^2)$$

So,

$$\begin{aligned} \langle x|e^{-\frac{\epsilon p^2}{2m}} e^{-\epsilon V(\hat{x})}|y\rangle & \quad \text{inserting } I = \int \frac{dp}{2\pi} |p\rangle\langle p| \\ &= \int \frac{dp}{2\pi} \langle x|e^{-\frac{\epsilon p^2}{2m}} |p\rangle\langle p|e^{-\epsilon V(\hat{x})}|y\rangle \\ &= \int \frac{dp}{2\pi} e^{-\frac{\epsilon p^2}{2m}} e^{-\epsilon V(x)} e^{ip \cdot (x-y)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon p^2}{2m} + ip \cdot (x-y)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon}{2m} \left(p^2 - \frac{2mi(x-y)}{\epsilon} p + \frac{4m^2(x-y)^2}{\epsilon^2} - \frac{4m^2(x-y)^2}{\epsilon^2} \right)} \\ &= \frac{1}{(2\pi)^2} e^{-\epsilon V(x)} \int dp e^{-\frac{\epsilon}{2m} \left(p - \frac{2mi(x-y)}{\epsilon} \right)^2} e^{-\frac{2m}{\epsilon} (x-y)^2} \\ &= \frac{1}{(2\pi)^2} e^{-(\epsilon V(x) + \frac{2m}{\epsilon} (x-y)^2)} \sqrt{\frac{\pi 2m}{\epsilon}} \end{aligned}$$

So,

$$\begin{aligned} Z &= \int dx_0 dx_1 \cdots dx_{N-1} \prod_{i=0}^{N-1} \frac{1}{(2\pi)^2} e^{-(\epsilon V(x_i) + \frac{2m}{\epsilon} (x_i - x_{i+1})^2)} \cdot \sqrt{\frac{2m\pi}{\epsilon}} \\ &= \frac{\sqrt{m}}{(2\pi)^{3/2}} \left(\frac{1}{\sqrt{\epsilon}} \right)^N \int dx_0 dx_1 \cdots dx_{N-1} e^{-\epsilon \left(\sum_{i=0}^{N-1} \left(V(x_i) + 2m \left(\frac{dx_i}{dt} \right)^2 \right) \right)} \\ & \quad \quad \quad \downarrow S_E \\ &= \sqrt{\frac{m}{\epsilon}} \left(\frac{1}{\sqrt{(2\pi)^3}} \right)^N \int \mathcal{D}x e^{-S_E} \end{aligned}$$

Note : Path integral approach is difficult / lengthy for Q.M. but for field theory this is much helpful approach.

For bosons we have commutator relations but for fermions we have anti commutation relations

$$\begin{aligned} \{a, a^\dagger\} = 1 & \quad \longrightarrow \quad 2a^\dagger a^\dagger = 0, \quad (a^\dagger)^2 = 0 \\ \{a, a\} = \{a^\dagger, a^\dagger\} &= 0 \end{aligned}$$

we also define fermionic vacuum $|0\rangle$ as

$$\begin{aligned} a|0\rangle &= 0 \\ a^\dagger|0\rangle &= \text{some state} \dots |1\rangle \\ (a^\dagger)(a^\dagger)|0\rangle &= 0|0\rangle = 0 \end{aligned}$$

So, $a^\dagger|1\rangle = a^\dagger a^\dagger|0\rangle = 0$

So, we have only two states $|0\rangle$ & $|1\rangle$.

if we try to add one more particle by $(a^\dagger)^2|0\rangle$ which becomes zero. (obeys Pauli-exclusion principle; does not allow 2 particles).

$$H = \frac{\hbar\omega}{2} (a^\dagger a - a a^\dagger) = \hbar\omega \left(a^\dagger a - \frac{1}{2} \right)$$

Such that,

$$H|0\rangle = -\frac{\hbar\omega}{2}|0\rangle$$

$$H|1\rangle = \frac{\hbar\omega}{2}|1\rangle$$

$$\text{So, } H = \frac{\hbar\omega}{2} \sigma_3 = \frac{\hbar\omega}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

we just have 2 eigen states of H .

Partition Fun :

$$\begin{aligned} Z &= \text{Tr} \left(e^{-\beta H} \right) = \underbrace{\langle 0 | e^{-\beta H} | 0 \rangle}_{\textcircled{1}} + \underbrace{\langle 1 | e^{-\beta H} | 1 \rangle}_{\textcircled{2}} \\ &= \langle 0 | 1 - \beta H + \frac{\beta^2 H^2}{2!} - \frac{\beta^3 H^3}{3!} + \dots | 0 \rangle + \langle 1 | 1 - \beta H + \frac{\beta^2 H^2}{2!} - \dots | 1 \rangle \\ \textcircled{1} &= \langle 0 | 0 \rangle - \beta \langle 0 | H | 0 \rangle + \frac{\beta^2}{2!} \langle 0 | H^2 | 0 \rangle - \dots \\ &= 1 + \beta \frac{\hbar\omega}{2} + \frac{\beta^2}{2!} \frac{\hbar^2 \omega^2}{4} + \dots = e^{\frac{\beta \hbar \omega}{2}} \end{aligned}$$

Similarly $\textcircled{2} = e^{-\frac{\beta \hbar \omega}{2}}$

$$Z = e^{\frac{\beta \hbar \omega}{2}} + e^{-\frac{\beta \hbar \omega}{2}} = 2 \cosh \frac{\beta \hbar \omega}{2}$$

Path Integrals for fermionic Z :

Grassmann variables c, c^*

Axioms :

(1) c & c^* are independent

(2) $c^2 = (c^*)^2 = 0$ $\{c, c^*\} = 0$

(3) Integrals –

$$(a) \int dc = 0 = \int dc^*$$

$$(b) \int dc \, c = 1 = \int dc^* \, c^*$$

$$(4) \quad 0 = \{c, dc\} = \{c^*, dc^*\} = \{c, dc^*\} = \{c^*, dc\}$$

$$\Rightarrow \int c dc = - \int dc c = -1$$

$$(5) \quad \{dc, dc\} = \{dc^*, dc^*\} = \{dc, dc^*\} = \{dc^*, dc\} = 0$$

Convention : write '*' variables to extreme left.

$$\int dc^* dc \quad \text{while doing mult}^n \quad - \text{Grassmann variable integrals.}$$

(6) How fermionic operators & Grassmann var. are related –

- 1) c, c^* anticommute with $a, a^\dagger$
- 2) $ca^\dagger$ (or) c^*a behave like ordinary operator
 $ex \quad [ca^\dagger, c^*] = 0$

Note : A state $|c\rangle$ which is labelled by Grassmann variable 'c' is $e^{-ca^\dagger}|0\rangle$

$$e^{-ca^\dagger}|0\rangle = (1 - ca^\dagger)|0\rangle$$

and,

$$\begin{aligned} a|c\rangle &= a e^{-ca^\dagger}|0\rangle \\ &= a (1 - ca^\dagger)|0\rangle \\ &= \cancel{a|0\rangle} \overset{0}{-} a c a^\dagger|0\rangle \\ &= - \underbrace{ac}_{\text{anticommute}} a^\dagger|0\rangle \\ &= + c a a^\dagger|0\rangle \\ a|c\rangle &= c|0\rangle \end{aligned} \quad (1)$$

also

$$\begin{aligned} c|c\rangle &= c (1 - ca^\dagger)|0\rangle \\ &= c|0\rangle - \cancel{c} \overset{0}{c} a^\dagger|0\rangle \\ &= c|0\rangle \end{aligned} \quad (2)$$

(1) & (2) implies $\longrightarrow$

$$\boxed{a|c\rangle = c|c\rangle}$$

5 Lecture 5

We were discussing Grassmann integrals & differentiation.

We discussed that $ca^\dagger, c^*a$ behaves like ordinary operators. Note the order matters.

$$[ca^\dagger, c^*] = 0$$

define $|c\rangle$ & $\langle c|$ as

$$|c\rangle = e^{-ca^\dagger}|0\rangle$$

We saw, $a|c\rangle = c|0\rangle = c|c\rangle$

&

$$\begin{aligned}
\langle c| &= \langle 0|e^{-ac^*} \\
&= \langle 0|(1 - ac^*) \\
&= \langle 0| - \langle 0|a c^* \\
&= \langle 0| - \langle 1|c^* \\
\langle c|a^\dagger &= \langle 0|a^\dagger - \langle 1|c^*a^\dagger \\
&= 0 + \langle 1|a^\dagger c^* \\
&= \langle 0|c^* \quad \text{-- verify!}
\end{aligned}$$

Overlap,

$$\begin{aligned}
\langle c'|c\rangle &= \langle 0|e^{-ac'^*} e^{-ca^\dagger}|0\rangle \\
&= \langle 0|(1 - ac'^*)(1 - ca^\dagger)|0\rangle \\
&= \langle 0|1 - ca^\dagger - ac'^* + ac'^*ca^\dagger|0\rangle \\
&= \langle 0|1 + a^\dagger c + c'^*a - c'^*a c a^\dagger|0\rangle \\
&= \langle 0|0\rangle + \cancel{\langle 0|a^\dagger c|0\rangle} + \cancel{\langle 0|c'^*a|0\rangle} + \langle 0|c'^*c a a^\dagger|0\rangle \\
\langle c'|c\rangle &= 1 + c'^*c \simeq e^{c'^*c}
\end{aligned}$$

Remember coherent states of SHO ($|z\rangle$)

$$I = C \int dz d\bar{z} e^{-\bar{z}z} |z\rangle\langle z|$$

Similarly

$$\begin{aligned}
I &= \int dc^* dc e^{-c^*c} |c\rangle\langle c| \\
&= \int dc^* dc (1 - c^*c) e^{-ca^\dagger}|0\rangle\langle 0|e^{-ac^*} \\
&= \int dc^* dc (1 - c^*c) (1 - ca^\dagger) |1\rangle\langle 0|(1 - ac^*) \\
&= \int dc^* dc (1 - c^*c) (|0\rangle - c|1\rangle) (\langle 0| - \langle 1|c^*) \\
&= \int dc^* dc (1 - c^*c) \left[|0\rangle\langle 0| - c^*|0\rangle\langle 1| - c|0\rangle\langle 0| \right. \\
&\quad \left. + c c^*|0\rangle\langle 1| \right] \\
&= \int dc^* dc (1 - c^*c) |0\rangle\langle 0| - \int dc^* dc (1 - c^*c) c^*|0\rangle\langle 1| \\
&\quad - \int dc^* dc (1 - c^*c) c|0\rangle\langle 0| + \int dc^* dc (1 - c^*c) c c^*|0\rangle\langle 1| \\
&= \cancel{\int dc^* dc |0\rangle\langle 0|} + \int dc^* dc c^*c |0\rangle\langle 0| - \cancel{\int dc^* dc |0\rangle\langle 1|} \\
&\quad + \int dc^* dc e^*c c^*|0\rangle\langle 1| - \cancel{\int dc^* dc c|0\rangle\langle 0|} + \int dc^* dc c c^*|0\rangle\langle 0| \\
&\quad + \int dc^* dc c c^*|0\rangle\langle 1| - \cancel{\int dc^* dc e^*c c c^*|0\rangle\langle 1|}
\end{aligned}$$

$$\begin{cases} c^2 = c^{*2} = 0 \\ \int dc \, c = \int dc^* \, c^* = 1 \\ \{c, c^*\} = 0 \end{cases}$$

We will get,

$$\underline{I} = |0\rangle\langle 0| + |1\rangle\langle 1|$$

Similarly, we can find

$$\begin{aligned} \text{Tr } A &= \langle 0|A|0\rangle + \langle 1|A|1\rangle \\ &= \int dc^* dc \, e^{-c^*c} \langle -c|A|c\rangle \\ &= \langle 0|A|0\rangle + \langle 1|A|1\rangle \end{aligned}$$

Find $Z = \text{Tr } e^{-\beta H}$

$$\begin{aligned} &= \int dc^* dc \, e^{-c^*c} \langle -c|e^{-\beta H}|c\rangle \quad \text{let } \epsilon = \frac{\beta}{N} \\ &= \int dc^* dc \, e^{-c^*c} \langle -c|e^{-\epsilon H} \cdot e^{-\epsilon H} \cdot e^{-\epsilon H} \dots e^{-\epsilon H}|c\rangle \end{aligned}$$

Insert

$$I = \int dc_i^* dc_i \, e^{-c_i^*c_i} |c_i\rangle\langle c_i|$$

We gonna need matrix elements like...

$$\begin{aligned} &e^{-c_{i+1}^*c_{i+1}} \langle c_{i+1}|e^{-\epsilon H(a, a^\dagger)}|c_i\rangle \\ &\simeq e^{-c_{i+1}^*c_{i+1}} \langle c_{i+1}|c_i\rangle e^{-\epsilon H(c_{i+1}^*, c_i)} \quad \left(\longrightarrow e^{c_{i+1}^*c_i}\right) \\ &= \exp \left[-\epsilon \left(c_{i+1}^* \frac{(c_{i+1} - c_i)}{\epsilon} - H(c_{i+1}^*, c_i) \right) \right] \\ Z &= \int dc_N^* dc_N \, dc_{N-1}^* dc_{N-1} \dots dc_1^* dc_1 \, e^{-S_E/\hbar} \end{aligned}$$

where,

$$\begin{aligned} S_E &= \epsilon \sum_{i=1}^N \left[c_{i+1}^* \left(\frac{c_{i+1} - c_i}{\epsilon} \right) + H(c_{i+1}^*, c_i) \right] \\ c_{N+1} &= -c_1 \\ c_{N+1}^* &= -c_1^* \end{aligned}$$

Finally we take $N \rightarrow \infty$, $\epsilon \rightarrow 0$ with $\beta = \epsilon N$

$$Z = \int \mathcal{D}c^*(\tau) \mathcal{D}c(\tau) \, e^{-\int_0^{\beta\hbar} d\tau \, (c^* \dot{c} - H)}$$

$$\left. \begin{aligned} c(\beta\hbar) &= -c(0) \\ c^*(\beta\hbar) &= -c^*(0) \end{aligned} \right\} \begin{array}{l} \text{Antiperiodic boundary cond}^n. \\ \text{in imaginary time axis or} \\ \text{axis. (inv. Temp)} \end{array}$$

Problem :

Consider $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + gx$

Solve by “completing the square”.

Also do variation of above

$$H = H_{SHO} + \underbrace{g(t)x}_{\text{position}} + \underbrace{p f(t)}_{\text{momentum}}$$

$$H = H_{SHO} + a F^*(t) + a^\dagger F(t) \quad \left(\downarrow \frac{f - ig}{\sqrt{2}} \right)$$

– compute the propagator

– compute the vacuum to vacuum transition amplitude.

$$\langle 0; t_{-\infty} | 0, t_{\infty} \rangle \quad \text{where } \{F, g \rightarrow 0 \text{ as } t \rightarrow \pm\infty\}$$

(margin working)

$$H' = a \left(\frac{f + ig}{\sqrt{2}} \right) + a^\dagger \left(\frac{f - ig}{\sqrt{2}} \right) \quad a = \frac{x + ip}{\sqrt{2}}, \quad a^\dagger = \frac{x - ip}{\sqrt{2}}$$

$$\left(\frac{x + ip}{\sqrt{2}} \right) \left(\frac{f + ig}{\sqrt{2}} \right) + \left(\frac{x - ip}{\sqrt{2}} \right) \left(\frac{f - ig}{\sqrt{2}} \right)$$

$$= \frac{1}{2} (xf + ixg + ipf - pg + xf - ixg - ipf - pg)$$

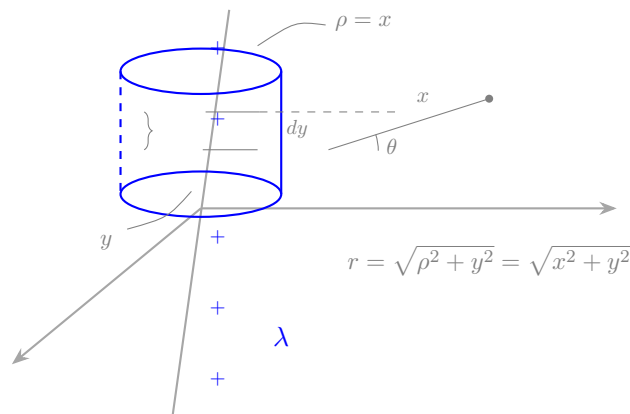
$$= \frac{1}{2} (x(f + ig + f - ig) + p(if - g - if - g))$$

$$= \frac{1}{2} (2xf - 2pg) = \underline{xf} - \underline{pg}.$$

Regularization And Renormalization :

These are not features of QFT alone.

ex : Electrostatics :



$$\phi = \frac{q}{\epsilon_0} \quad \Rightarrow \quad E(2\pi x \ell) = \frac{\lambda \ell}{\epsilon_0}$$

$$E(x) = \frac{\lambda}{2\pi x \epsilon_0}$$

$$\phi(x) - \phi(y) = \int_x^y \vec{E} \cdot d\vec{x}$$

using $E(x)$.

$$\phi(x) - \phi(y) = \frac{\lambda}{2\pi \epsilon_0} \ln \frac{y}{x}$$

But we know

$$\phi(r) = \int \frac{k dq}{r} \quad (\text{here } x = \rho)$$

$$= \frac{1}{4\pi \epsilon_0} \int_{-\infty}^{\infty} \frac{\lambda dy}{\sqrt{x^2 + y^2}}$$

$$\left. \begin{array}{l} y = x \tan \theta \\ dy = x \sec^2 \theta d\theta \end{array} \right\} \Rightarrow \phi(r) = k\lambda \int \frac{x \sec^2 \theta d\theta}{x \sec \theta} = k\lambda \int \sec \theta d\theta$$

$$\Rightarrow \phi(r) = k\lambda \int \frac{\sec \theta (\sec \theta + \tan \theta) d\theta}{\sec \theta + \tan \theta}$$

$$\left. \begin{array}{l} \sec \theta + \tan \theta = t \\ (\sec \theta \tan \theta + \sec^2 \theta) d\theta = dt \end{array} \right\} \quad \phi(r) = k\lambda \int \frac{dt}{t} = k\lambda \ln t = k\lambda \ln (\sec \theta + \tan \theta) \Big|_{-\pi/2}^{\pi/2}$$

$$= \infty - \infty$$

But answer shall be finite!

Similarly when we calculate things in QFT, we sometimes get ∞ , but we know output must be some finite no.

put $\eta = y/x_0$

$$\phi(x) - \phi(x_0) = \frac{\lambda}{4\pi \epsilon_0} \int du \left[\left(\left(\frac{x}{x_0} \right)^2 + u^2 \right)^{-1/2} - (1 + u^2)^{-1/2} \right]$$

use,

$$-\frac{1}{2} \int_1^b d\alpha (u^2 + \alpha)^{-3/2} = (u^2 + b)^{-1/2} - (u^2 + 1)^{-1/2}$$

$$\phi(x) - \phi(x_0) = \frac{-\lambda}{8\pi \epsilon_0} \int_{-\infty}^{\infty} du \int_1^{(x/x_0)^2} d\alpha (u^2 + \alpha)^{-3/2}$$

put,

$$z = u \alpha^{-1/2} (x/x_0)^2$$

$$\begin{aligned} \phi(x) - \phi(x_0) &= -\frac{\lambda}{8\pi \epsilon_0} \int_1^{(x/x_0)^2} \frac{d\alpha}{\alpha} \int_{-\infty}^{\infty} \frac{dz}{(1 + z^2)^{3/2}} \\ &= \frac{-\lambda}{2\pi \epsilon_0} \ln \frac{x}{x_0} \end{aligned}$$

Above way is called dimensional Regularization (DR). DR is faster.

Now we will perform same calculation with DR :

Notice that $\phi(x)$ is scale invariant. $\begin{cases} x \rightarrow kx \\ y \rightarrow ky \end{cases}$

$$\phi(kx) = \phi(x)$$

$$\phi(kx) = \int_{-\infty}^{\infty} \frac{k dy}{k \sqrt{y^2 + x^2}} = \underbrace{\int_{-\infty}^{\infty} \frac{dy}{\sqrt{y^2 + x^2}}}_{\text{This integral is divergent}} = \phi(x)$$

For d -dimension ϕ is not scale invariant!

$$\phi_d(x) = \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \int_{-\infty}^{\infty} \frac{d^d y}{\sqrt{y^2 + x^2}}$$

$\downarrow d$ – is dimension of charge confⁿ.

μ has dimension of $\frac{1}{\ell}$ i.e. μ has mass dimension.

Because

$$\phi_d(kx) = k^{d-1} \phi(x)$$

to make it scale invariant we multiply by μ^{d-1} . ($\mu = \text{const.}$)

$$\phi_d(x) = \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \int \frac{\rho^{d-1} d\rho}{\sqrt{\rho^2 + x^2}} \int d\Omega_{d-1}$$

i.e. we will have $d-1$ angular variables & 1 radial variable, in d -dimensions.

$$= \frac{\lambda}{4\pi\epsilon_0} \mu^{d-1} \frac{2(\pi)^{d/2}}{\Gamma(d/2)} \underbrace{\int_0^{\infty} \frac{\rho^{d-1} d\rho}{(\rho^2 + x^2)^{1/2}}}_{I} \quad \text{check!}$$

$$I = \frac{\frac{1}{2} \Gamma(d/2) \Gamma(\frac{1-d}{2})}{\sqrt{\pi}} \quad \leftarrow \text{should include 'x'!}$$

$$(z-1)! = \Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt \quad m! = \Gamma(m+1) = \int_0^{\infty} t^m (e^{-t} dt)$$

6 Lecture 6

Gamma fun. is defined for $\Gamma(x)$, $x > 0$.

for $\Gamma\left(\frac{1-d}{2}\right)$ for $\frac{1-d}{2} < 0$ we get poles.

$$\Gamma_m = (m-1)! = \int_0^{\infty} t^{m-1} e^{-t} dt \quad \leftrightarrow \text{has poles when } m < 0. \quad \text{Similarly } \Gamma\left(\frac{1-d}{2}\right) \text{ has poles for } \frac{1-d}{2} < 1 \Rightarrow 1-d < 2 \Rightarrow d > -1$$

↑ we are extending domain of d from +ve integer to real numbers, by analytic continuation.

Expand ϕ_d in powers of $1 - d \equiv \epsilon$

$$\begin{aligned} \text{if } \frac{1-d}{2} &= -\text{ve integer} \\ 1-d &= -\text{ve even integer} \\ \Rightarrow d &= +\text{ve odd integer} \end{aligned}$$

Note that we can't add quantities of different dimensions.

So $\exp(\text{dimensionful quantity})$ = does not make sense.

μ^{d-1}, x^{d-1} if expanded separately then above problem arise.

So, we combine

$$\begin{aligned} (\mu x)^{d-1} &= (\mu x)^{-\epsilon} \\ &= 1 - \epsilon \ln \mu x + O(\epsilon^2) \quad > \text{should be minus!} \end{aligned}$$

$$x^n = e^{\ln x^n}; \quad (\mu x)^{-\epsilon} = e^{\ln(\mu x)^{-\epsilon}} = 1 + \ln(\mu x)^{-\epsilon} + \frac{(\ln(\mu x)^{-\epsilon})^2}{2!} + \dots = 1 - \epsilon \ln \mu x + O(\epsilon)^2$$

$$\pi^{\frac{d-1}{2}} = 1 - \frac{\epsilon}{2} \ln \pi + O(\epsilon^2)$$

$$\Gamma\left(\frac{1-d}{2}\right) = \Gamma(-\epsilon/2) = \frac{-2}{\epsilon} \Gamma(1 - \epsilon/2) \quad (?)$$

$$\Gamma(z+1) = z \Gamma(z); \quad \Gamma(-\epsilon/2) = \left(-\frac{\epsilon}{2}\right)^{-1} \Gamma\left(-\frac{\epsilon}{2} + 1\right) = \left(-\frac{2}{\epsilon}\right) \Gamma\left(1 - \frac{\epsilon}{2}\right)$$

$$= \frac{-2}{\epsilon} \left(1 + \frac{\epsilon \gamma}{2} + O(\epsilon^2) + \dots\right)$$

So,

$$\phi_d(x) = \frac{-\lambda}{2\pi\epsilon_0} \left[\frac{1}{\epsilon} + \underbrace{\frac{1}{2} \ln \pi + \frac{\gamma}{2} + \ln \mu x}_{\ln \mu' x} + O(\epsilon) \right]$$

↓ Can be dropped for $d = 1 \Rightarrow \underline{\epsilon \equiv 0}$.

define

$$\phi_{d=1}^d(x) = \frac{-\lambda}{2\pi\epsilon_0} (\ln \mu' x)$$

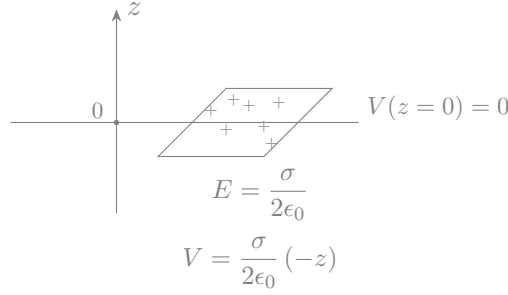
$$\phi^R(x) = \frac{A}{\epsilon} + \left(\frac{-\lambda}{2\pi\epsilon_0}\right) \ln \mu' x$$

$$\phi^R(y) = \frac{A}{\epsilon} + \left(\frac{-\lambda}{2\pi\epsilon_0}\right) \ln \mu' x'$$

$$\phi^R(x) - \phi^R(x') = \frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{x}{x'}\right) \quad \leftarrow \text{It is potential difference that are finite!}$$

We have found this answer from basic defⁿ of potential $V(r) = \int \frac{k dq}{r}$

For 2d charge confⁿ.



Generating Functional for interacting scalar field :

$$Z[J] = e^{\frac{1}{2}\langle JGJ \rangle} \left[e^{\langle JG\delta \rangle} \mathcal{F}[\varphi] \right]_{\varphi=0}$$

Where,

$$\langle JGJ \rangle = \int d^4x d^4y J(x) G(x, y) J(y) \quad \text{Check } \varphi^4 d^4x ?$$

$$\mathcal{F}[\varphi] = N e^{\frac{1}{2}\langle \delta G \delta \rangle} e^{-\frac{i\lambda}{4!} \int \varphi^4 dx}$$

$$Z_0[J] = e^{\frac{1}{2}\langle JGJ \rangle}$$

So,

$$Z[J] = N \left[e^{\frac{i\lambda}{4!} \int d^4x \left(\frac{\delta}{\delta J(x)} \right)^4} \right] Z_0[J]$$

Gaussian functional integral

↓

Greens fun are functional integrals

↓

$Z_0[J]$ as a functional integral

↓

Z as a functional integral.

$$Z[J] = N \int [d\varphi] e^{iS - \epsilon \int \varphi^2 + J\varphi}$$

$$S \rightarrow S_0 - \frac{\lambda}{4!} \int d^4x \varphi^4(x) \quad \begin{cases} S = S_M \\ S_E = -iS_M \end{cases}$$

The gaussian integral is of type

$$\int dx e^{i\alpha x^2 - \epsilon x^2} = \sqrt{\frac{\pi}{\epsilon - i\alpha}}$$

We can equally work with euclidean action.

$$t \rightarrow -i\tau, \quad S_E = -iS_M \quad \text{here, } \tau = \text{imag}, t = \text{real}$$

$$Z[J] \sim \int [d\varphi_E] e^{-S_E}$$

$$S_E = \int d^4x \left(\frac{(\partial\varphi)^2}{2} + \frac{\mu^2\varphi^2}{2} + \frac{\lambda\varphi^4}{4} \right)$$

Renormalisation (why in QFT) :

1. loop integrals involve integrals over arbitrary large momenta.
2. Theory (i.e. our $\mathcal{L}$) is not valid at extremely high energies / momenta. (We impose a cut off on momenta, then integrals will be dependent on momentum cut off (say Λ). (known as Λ -sensitive integrals for a given diagram.)
 $\Rightarrow$ Then loop integrals become finite but depend on Λ .

This is like cutting infinite wire to make it wire of finite length ℓ .

where, $\ell \rightarrow \infty$ is approximation of real world.

$$\ell [\underline{V \sim \ln r} \quad \longrightarrow \quad r' \quad \left(V(r') \neq \sim \ln(r) \sim \frac{1}{r} \right)$$

Transition from $\ln(r) \longrightarrow \frac{1}{r}$.

3. Scattering amplitudes depend on parameters of $\mathcal{L}$ (like $\lambda_0, \mu_0 \dots$) $\leftrightarrow$ Bare Coupling const. & the actual measured couplings $(\lambda, \mu \dots)$ are functions of λ_0, μ_0 and Λ (momentum cut off).

$$\lambda = f(\lambda_0, \mu_0, \Lambda) ; \quad \mu = g(\lambda_0, \mu_0, \Lambda)$$

What ? { The Λ -dependence of calculated results can be absorbed between μ_0, λ_0 and $\mu, \lambda \dots$

$\longrightarrow$ let us see this for φ^4 theory.

Start with,

$$\begin{aligned} \mathcal{L}_M &= \frac{1}{2} \alpha^2 (\partial\varphi)^2 - \frac{\mu^2 \varphi^2}{2} - \frac{\lambda}{4} \varphi^4 - J\varphi \\ \mathcal{L}_E &= \frac{1}{2} \alpha^2 (\partial\varphi)^2 + \frac{\mu^2 \varphi^2}{2} + \frac{\lambda}{4} \varphi^4 + J\varphi \end{aligned}$$

So,

$$Z[J] = \int [d\varphi] e^{-S_E + J\varphi} \quad (\text{Meaning of source term ?})$$

$\downarrow$ generates correlation functions / greens fun. which gives scattering amplitudes.

Correlation functions

$$G^{(n)}(x_1, x_2 \dots x_n; J) = \frac{1}{Z[J]} \frac{\delta}{\delta J(x_1)} \frac{\delta}{\delta J(x_2)} \dots \frac{\delta}{\delta J(x_n)} Z[J]$$

$\longleftarrow$ includes disconnected diagram as well

Connected correlation functions. (How ?) $\rightarrow$ will derive tomorrow!

$$G_c^{(n)}(x_1, x_2 \dots x_n; J) = \frac{\delta}{\delta J(x_1)} \dots \frac{\delta}{\delta J(x_n)} \ln Z[J]$$

Mass Renormalisation :

(Srednicki S.24)

Quantity of interest is 2-point vertex function

$$\begin{aligned}\Gamma^{(2)}(k) &= \left(\text{---} \xrightarrow{k} \text{---} \right)^{-1} + \frac{\text{---} \bigcirc^q \text{---}}{k} + O(\lambda^2) \\ &= (\alpha^2 k^2 + \mu^2) + \frac{\lambda}{2} \int^\Lambda \frac{d^4 q}{(\alpha^2 q^2 + \mu^2) (2\pi)^4} + O(\lambda^2) \quad - \quad \textcircled{1}\end{aligned}$$

$\Lambda \leftarrow$ cut off

We demand that, $\Gamma^{(2)}(k=0)$ be equal to (physical mass)².

$$\begin{aligned}\Gamma^{(2)}(k) \Big|_{k=0} &= m^2 \quad \left(\tilde{q} = \frac{q}{2\pi} \right) \\ \Gamma^{(2)}(0) &= m^2 = \alpha^2(0) + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{d^4 \tilde{q}}{(\alpha^2 q^2 + \mu^2)} + O(\lambda^2) \\ \Gamma^{(2)}(k) &= \Gamma^{(2)}(k) + 0 \\ &= \Gamma^{(2)}(k) - \Gamma^{(2)}(0) + m^2 \\ &= \alpha^2 k^2 + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{\cancel{d^4 q}}{(2\pi)^4} \frac{1}{(\alpha^2 q^2 + \mu^2)} + O(\lambda^2) \\ &\quad - \left(0 + \mu^2 + \frac{\lambda}{2} \int_0^\Lambda \frac{\cancel{d^4 \tilde{q}}}{(\alpha^2 \tilde{q}^2 + \mu^2)} + O(\lambda^2) \right) \\ &\quad + m^2 \\ &\quad \boxed{\Gamma^{(2)}(k) = \alpha^2 k^2 + m^2 + O(\lambda^2)}\end{aligned}$$

7 Lecture 7 (30 Jan 2023)

Reading Assignment – 2 Read Srednicki chapters 12–20. and solve all problems.

Mass Renormalisation –

$\Gamma^{(2)}(k)$ depends on Λ only then $\exists$ relation b/w μ & m .

$$\int \frac{d^4 \tilde{q}}{\alpha^2 q^2 + \mu^2} = \frac{V(S^{d-1})}{(2\pi)^d} \int_0^\Lambda \frac{q^{d-1} dq}{\alpha^2 q^2 + \mu^2} ; \quad \text{where } \left\{ d^d \tilde{q} = \frac{d^d q}{(2\pi)^d} \right.$$

For $d = 4$;

$$\frac{V(S^3)}{(2\pi)^4} \int_0^\Lambda \frac{q^3 dq}{\alpha^2 q^2 + \mu^2} = \left(\alpha^2 \Lambda^2 + \mu^2 \ln \mu^2 - \underbrace{\mu^2 \ln (\mu^2 + \alpha^2 \Lambda^2)}_{\downarrow} \right)$$

Subleading term.

(margin working)

$$\begin{aligned}
\int \frac{dq \, q^3}{\alpha^2 q^2 + \mu^2} &= \frac{1}{\alpha^2} \int_0^\Lambda \frac{dq (\alpha^2 q^2 + \mu^2 - \mu^2) q}{\alpha^2 q^2 + \mu^2} & \alpha^2 q^2 + \mu^2 &= t \\
&= \frac{1}{2\alpha^4} \int \frac{dt (t - \mu^2)}{t} & 2\alpha^2 q \, dq &= dt \\
&= \frac{1}{2\alpha^4} \int dt \left(1 - \frac{\mu^2}{t} \right) = \frac{1}{2\alpha^4} (t - \mu^2 \ln t) \\
&= \frac{1}{2\alpha^4} (\alpha^2 q^2 + \mu^2 - \mu^2 \ln (\alpha^2 q^2 + \mu^2)) \Big|_0^\Lambda \\
&= \frac{1}{2\alpha^4} (\alpha^2 \Lambda^2 + \mu^2 - \mu^2 \ln (\alpha^2 \Lambda^2 + \mu^2) - (\mu^2 - \mu^2 \ln \mu^2)) \\
&= \frac{1}{2\alpha^4} \left(\alpha^2 \Lambda^2 + \cancel{\mu^2} - \mu^2 \ln \left(\frac{\alpha^2 \Lambda^2 + \mu^2}{\mu^2} \right) \right) \\
&= \frac{1}{2\alpha^4} \left(\alpha^2 \Lambda^2 - \mu^2 \ln \left(\frac{\alpha^2 \Lambda^2}{\mu^2} + 1 \right) \right)
\end{aligned}$$

From eqⁿ ① of 6th lecture –

$$m^2 = \mu^2 + \frac{\lambda}{2} \left(\alpha^2 \Lambda^2 + \mu^2 \ln \left(\frac{\mu^2}{\mu^2 + \alpha^2 \Lambda^2} \right) \right)$$

$$\text{This gives } \mu^2 = \mu^2(\Lambda; m^2)$$

look at 2 loops –

$$\begin{aligned}
\Gamma^{(2)} &= (\text{tree level} + 1 \text{ loop}) + \text{diagram with two bubbles} \\
&+ \text{diagram with a bubble}
\end{aligned}$$

$$\begin{aligned}
&= \mu^2 + \alpha^2 k^2 + \frac{1}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + \mu^2} - \lambda^2 \left(\frac{1}{4} \int \frac{d^d \tilde{q}_1 \, d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2)} \right. \\
&\quad \left. + \frac{1}{6} \int \frac{d^d \tilde{q}_1 \, d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2) (\alpha^2 (k - Q)^2 + \mu^2)} \right) + O(\lambda^3)
\end{aligned}$$

Again define $\Gamma^{(2)}(k=0) = m^2$ and write

$$\Gamma^{(2)}(k) = \Gamma^{(2)}(k) - \Gamma^{(2)}(0) + m^2 + O(\lambda^3)$$

$$\Gamma^{(2)}(k) = \alpha^2 k^2 + m^2 - \frac{\lambda^2}{6} \Delta A(k)$$

$$\Delta A(k) = A(k) - A(0) \quad Q = \tilde{q}_1 + \tilde{q}_2$$

$$\begin{aligned}\Delta A(k) &= \int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2)(\alpha^2 q_2^2 + \mu^2)} \left(\frac{1}{\alpha^2 (k-Q)^2 + \mu^2} - \frac{1}{\alpha^2 Q^2 + \mu^2} \right) \\ &= \int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha^2 q_1^2 + \mu^2)(\alpha^2 q_2^2 + \mu^2)} \left(\frac{\alpha^2 Q^2 - \alpha^2 (k-Q)^2}{(\alpha^2 (k-Q)^2 + \mu^2)(\alpha^2 Q^2 + \mu^2)} \right)\end{aligned}$$

It is divergent for $(2d - 4 + 1 - 4 = 2d - 7)$ (?)

λ should be small so that, it nullifies effect of large Λ when series expanded in λ . Why so much small ?

Eliminate μ in favour of $m \rightarrow$ replace $\mu \leftrightarrow m$ in RHS; (?)

$$\begin{aligned}\Gamma^{(2)}(k) &= m^2 + \alpha^2 k^2 - \frac{\lambda^2}{6} \Delta A(k) \\ \mu^2 &= m^2 - \frac{\lambda}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + m^2} + \lambda^2 \left[\int \frac{d^d \tilde{q}_1 d^d \tilde{q}_2}{(\alpha_1^2 q_1^2 + m^2)(\alpha_2^2 q_2^2 + m^2)} + \frac{1}{6} A(0) \right] + O(\lambda^3)\end{aligned}$$

We have one more relation available to re-define

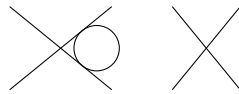
$$\begin{aligned}a^2 &\equiv \left. \frac{d\Gamma^{(2)}(k)}{dk^2} \right|_{k=\lambda} \\ \Gamma^{(2)}(k) &= \Gamma^{(2)}(k) - k^2 \left. \frac{d\Gamma^{(2)}}{dk^2} \right|_k + k^2 \frac{d\Gamma^{(2)}}{dk^2} \\ &= m^2 + \alpha^2 k^2 - \frac{\lambda^2}{6} (\Delta A(k)) - k^2 \left. \frac{d\Gamma^{(2)}}{dk^2} \right|_k \\ &\quad + k^2 a^2 + O(\lambda^2) \\ &= m^2 + k^2 \left(a^2 - \frac{\lambda^2}{6} B(k, \lambda) \right) + O(\lambda^3) \\ \downarrow &\quad \frac{\Delta A(k)}{k^2} + \frac{6}{\lambda^2} \left(\frac{d\Gamma^{(2)}}{dk^2} \right)_k - \alpha^2\end{aligned}$$

So we know B ;

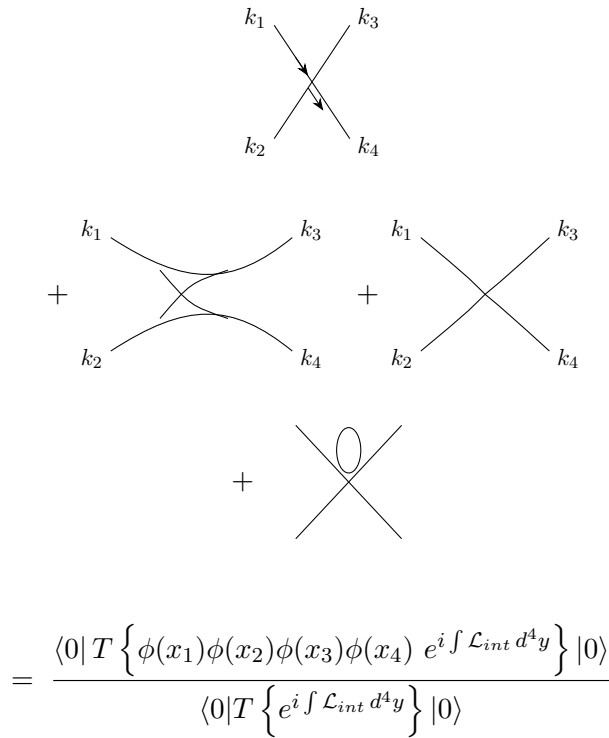
$$\frac{d\Gamma^{(2)}}{dk^2} = \text{[?]} = -\alpha \quad (\text{in free theory})$$

Renormalisation of λ :-

Eliminate λ in favour of $g \equiv \Gamma^{(4)}(0, 0, 0)$



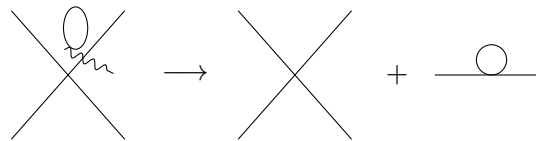
Proceed as earlier :-



$$= \frac{\langle 0 | T \left\{ \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) e^{i \int \mathcal{L}_{int} d^4 y} \right\} | 0 \rangle}{\langle 0 | T \left\{ e^{i \int \mathcal{L}_{int} d^4 y} \right\} | 0 \rangle}$$

disconnected diagrams cancels away because of $\log[Z]$ term in path integral.

Path integrals shouldn't be worried about because they fall apart into 2 diagrams.



$$\text{Diagram with loop on contact} \rightarrow \text{Contact diagram} + \text{Diagram with loop on external line}$$

$$\Gamma^{(2)}(k_1, k_2, k_3) = \lambda - \frac{\lambda^2}{2} \int \frac{d^d \tilde{q}}{\alpha^2 q^2 + m^2} \left(\frac{1}{a^2 (k - q)^2 + m^2} + \frac{1}{a^2 (k_1 + k_3 - q)^2 + m^2} + \frac{1}{a^2 (k_1 + k_3 - q)^2 + m^2} \right)$$

Add and subtract $g = \Gamma^{(4)}(0, 0, 0)$

$$\Gamma^{(4)}(k_1, k_2, k_3) = \Gamma^{(4)}(k_1, k_2, k_3) - \Gamma^{(4)}(0, 0, 0)$$

8 Lecture 8

Infrared cut off — means low energy cut off
ultraviolet cut off means high " " off.

$$m^2 = \mu^2 + \frac{1}{2} \left(\alpha^2 \Lambda^2 + \mu^2 \ln \frac{\mu^2}{\mu^2 + \alpha^2 \Lambda^2} \right)$$

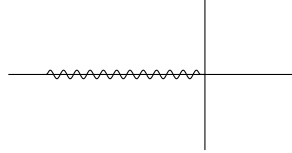
$$Z(m, \lambda) = \int [d\phi] e^{-\frac{1}{2} \int d^4 x ((\partial\phi)^2 + m^2 \phi^2 + \lambda \phi^4)}$$

$$= Z_0(m, 0) + \lambda Z_1(\quad) + \lambda^2 Z_2(\quad) + \dots$$

We know,

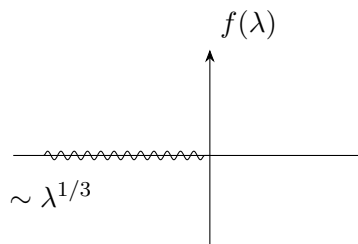
$$I(a) = \int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}} \sim a^{-1/2}$$

Branch cut starts at $a = 0$



$$\begin{aligned} I(a, \lambda) &= \int_{-\infty}^{\infty} dx e^{-ax^2 - \lambda x^4/4} \\ I_\lambda(a) &= \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-\lambda x^4/4!} \\ &= \int_{-\infty}^{\infty} dx e^{-ax^2} \left(1 - \frac{\lambda x^4}{4!} + \frac{1}{2!} \frac{\lambda x^8}{4! 4!} - \dots \right) \\ I(a; m) &= \int_{-\infty}^{\infty} dx e^{-ax^2} x^m \\ \int_{-\infty}^{\infty} dx x^2 e^{-ax^2} &= -\frac{d}{da} I_0(a) \\ I_\lambda(a) &= I(a, 0) + \lambda I(a, 1) + \frac{\lambda^2}{2!} I(a, 2) + \dots \end{aligned}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \text{Not analytic} & \sim \frac{1}{\sqrt{a}} & \frac{1}{a^{5/2}} & \frac{1}{a^{7/2}} \\ \text{at } \lambda = 0 & & & \end{array}$$



Power series will only work iff λ is very small.
Asymptotic series works at $\lambda = 0$.

HELING (arxiv)

↓

To understand renormalisation & fall in love with it. (could not find)

Renormalisation of λ :

$$\Gamma^{(4)}(k_1, k_2, k_3) =$$

Eliminate λ in favour of $g \equiv \Gamma^{(4)}(0, 0, 0)$

Replace μ^2 by m^2 , α^2 by a^2 ..

$$\Gamma^{(4)}(\{k\}) = \Gamma^{(4)}(\{k_i\}) - \Gamma^{(4)}(\{0\}) + g$$

Renormalisation condⁿ :

$$g = \lambda - \frac{\lambda^2}{2} \int \frac{d^d \tilde{q}}{(a^2 q^2 + m^2)} \quad (3 \text{ terms})$$

$$\left. \begin{array}{l} y = x - ax^2 \\ x = y - ay^2 \end{array} \right\} \quad (\text{for small } x)$$

$$\Gamma^{(4)}(\{k_i\}) = g + \frac{g^2}{2} \left(\int \cdots \right)$$

We choose $(k_1, k_2, k_3) = (0, 0, 0)$ as point to define ‘ g ’

we could have used other possibilities

$$g(k_1, k_2, k_3) = (k_1 \ k_2 \ k_3)$$

A common choice —

Symmetric pt. — all k_i are equal and have magnitude $\sqrt{\frac{3}{2}} k$

$$\Gamma^{(2)} = m^2 + k^2 \left(a^2 - \frac{g^2}{6} B(k, K) \right) + O(g^3)$$

$$\Gamma^{(4)} = g + \frac{g^2}{2} \left(\int () \cdots \right) + O(g^3)$$

$$\lambda = g \left(1 + \frac{3}{2} g \int \frac{d^d \tilde{q}}{(\alpha^2 q^2 + m^2)} \right) + O(g^3)$$

$$\mu^2 = m^2 - \frac{\lambda}{2} \int \frac{d^d \tilde{q}}{(\alpha^2 q^2 + m^2)} + \frac{\lambda^2}{6} A(0)$$

↘ substitute $g = g(\lambda)$

(Q) What about higher loops? What is minimum number of Λ -sensitive integrals?

We have 3 of them.

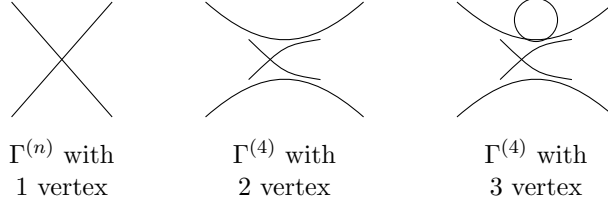
$$\begin{aligned}\mu^2 &= \mu^2(m^2, \lambda, \Lambda) \\ \lambda^2 &= \lambda^2(m^2, g, \Lambda) \\ \alpha^2 &= \alpha^2(m^2, \lambda, \Lambda)\end{aligned}$$

This is same as no. of bare parameters in L . So we have one parameter family of free parameters in L .

$$1 \text{ --- parameter family of lagrangians.} \\ (\alpha, \mu, \lambda, \Lambda)$$

all of which generates same renormalised theory, given by (a, m^2, g) .

Q) In general what is Λ -sensitivity for $\Gamma^{(n)}$?



Say $\Gamma^{(n)}$ has ' V ' vertices. Then internal lines

$$I = \frac{1}{2} (4V - n)$$

$$\text{No. of momenta} = n + \frac{1}{2} (4V - n)$$

$$\text{No. of individual momenta} = n + \frac{1}{2} (4V - n) - V$$

where $(n - 1)$ of these are independent external momenta.

$$(\text{No. of loop momenta} \equiv L) = V + \frac{n}{2} - (n - 1)$$

$$L = V - \frac{n}{2} + 1$$

In $\Gamma^{(n)}$ computation ; we will have integrals like

$$\frac{d^d \tilde{q}_1 d^d \tilde{q}_2 \cdots d^d \tilde{q}_L}{(\alpha^2 q_1^2 + \mu^2) (\alpha^2 q_2^2 + \mu^2) \cdots (\alpha^2 q_L^2 + \mu^2)}$$

goes like $q^{Ld-2L} \sim q^{(d-2)L} \sim q^{(d-2)(V-\frac{n}{2}+1)}$

look at large ' q ' behaviour —

For ex: $d = 4$ Behaviour of integral $\sim q^{2(V-\frac{n}{2}+1)}$

$$\sim q^{4-n} \quad \forall \text{ graphs of } \Gamma^{(n)}$$

$$\Gamma^{(2)} \sim q^2 \sim \Lambda^2$$

$$\Gamma^{(4)} \sim \ln \Lambda$$

$$\Gamma^{(n)} \sim (n > 4) \Rightarrow \text{not } \Lambda \text{ sensitive}$$

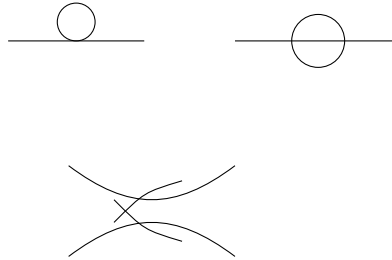
↘ Naive counting

1 P I (Particle irreducible)

These diagrams, cutting internal lines once will not be able to disconnect the diagrams / external lines.

A primitive divergent graph is one which is Λ -sensitive but has no Λ -sensitive ‘Sunset / sunrise / saturn’

only primitively
divergent
graphs.
($d \leq 4$)

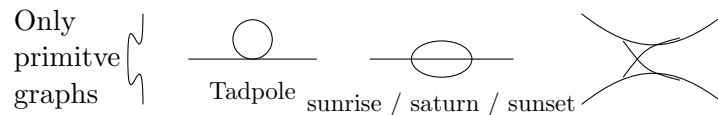


Every diagram ($\Gamma^{(n)}$; $n > 4$) is built out of $\Gamma^{(2)}$ & $\Gamma^{(4)}$.

9 Lecture 9

$$\text{Mid Term Exam} \begin{cases} 18^{\text{th}} \text{ or } 25^{\text{th}} \text{ Feb.} \\ 4 \text{ hours} \\ 100 \text{ points.} \end{cases}$$

We were talking about a graph that is Λ -sensitive has no Λ -sensitive subgraphs is called a primitively divergent graphs.

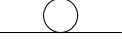


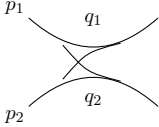
(in $\lambda\phi^4$ interaction, $d = 4$) (every other diagram is built out of $\Gamma^{(2)}$ & $\Gamma^{(4)}$)

Dimensional Regularisation — (D R)

$$\int_0^\Lambda \frac{q^3 dq}{q^2 + \mu^2} \sim \int_0^\Lambda \frac{q^3 dq}{q^2} \sim \int_0^\Lambda q dq \sim \Lambda^2$$

let's look at graph one by one.

1)  (Tadpole) : $\int \frac{d^4 \tilde{q}}{(q^2 + \mu^2)} \sim \Lambda^2$ quadratic UV-divergence

2)  : $\frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}_1 d^4 \tilde{q}_2}{(q_1^2 + \mu^2)(q_2^2 + \mu^2)} \delta(q_1 + q_2 - p)$

$$\sim \frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}}{(q_1^2 + \mu^2)((p - q_1)^2 + \mu^2)}$$

$$\longrightarrow \frac{\lambda^2}{2} \int \frac{d^4 \tilde{q}}{q^4} \sim \lambda^2 \ln \Lambda$$

If $\mu^2 = 0$; & $(p_1 + p_2)^2 = 0$, then there is also divergence for $q \rightarrow 0$ (IR divergence)

Typical integrals —

$$I_4(k) = \int d^4 \ell F(k, \ell)$$

For large $\ell \longrightarrow$ ℓ^{-2} for $\Gamma^{(2)}$
 ℓ^{-4} for $\Gamma^{(4)}$

For $d = 2$; $\int \frac{d^2 \ell}{\ell^4}$ is convergent in uv.

Suggest the following math. trick / procedure —

Consider, $I(w, k) = \int (d^{2w} \ell) F(\ell, k)$ ($2w = \text{dimension}$)

Treat 'w' as complex variable. (We are extending $\int d^d q$ integral to complex plane, via analytic continuation) $\overline{n+1} = n!$

We evaluate this integral in a region (of w-space), where there are no singularities. We then invent I' (new integral) which is convergent over bigger region / domain. This process is called analytic continuation in complex analysis.

$$\int d^{2w} \ell \boxed{F(\ell, k)} \quad \text{converges for } w < 2 \quad \left(\sim \frac{1}{\ell^2} \cdot \frac{1}{\ell^2} \right)$$

For our divergent integrals, this region is $w < 2$. We write :

$$\int d^{2w} \ell = \underbrace{\int d^4 \ell}_{\text{radial}} \underbrace{\int d^{2w-4} \ell}_{\text{angular}}$$

$$I = \int d^4\ell \underbrace{\int d\Omega_{2w-5}}_{\text{easy}} \int_0^\infty dL \frac{L^{2w-5}}{(L^2 + \ell^2 + \mu^2)}$$

$$\int_{-\infty}^\infty dx_1 \cdots \int_{-\infty}^\infty dx_N e^{-a(x_1^2 + x_2^2 + \cdots + x_N^2)} = \left(\frac{\pi}{a}\right)^{N/2}$$

Changing to polar coordinates

$$\underbrace{\int d\Omega_{N-1}}_{V(S^{N-1})} \int r^{N-1} dr e^{-ar^2} \quad (\sum x_i^2 = r^2)$$

↓ Volume of sphere of $(N-1)$ dim.

$$\Rightarrow V(S^{N-1}) \int r^{N-1} e^{-ar^2} dr \quad \begin{array}{l} ar^2 = t \\ ar dr = dt/2 \end{array}$$

$$V(S^{N-1}) \int \left(\frac{t}{a}\right)^{\frac{N-1}{2}} e^{-t} \frac{dt}{2a \left(\frac{t}{a}\right)^{1/2}}$$

$$\frac{1}{2a} V(S^{N-1}) \int \left(\frac{t}{a}\right)^{\frac{N-2}{2}} e^{-t} dt = \left(\frac{\pi}{a}\right)^{N/2}$$

$$\frac{1}{2a^{N/2}} V(S^{N-1}) \underbrace{\int t^{\frac{N}{2}-1} e^{-t} dt}$$

$$\frac{1}{2a^{N/2}} V(S^{N-1}) \Gamma(N/2) = \left(\frac{\pi}{a}\right)^{N/2}$$

$$\boxed{V(S^{N-1}) = \frac{2}{\Gamma(N/2)} \pi^{N/2}}$$

check: For $N=3$

$$V(S^2) = \frac{2\pi^{3/2}}{\Gamma(3/2)} = \frac{2\pi^{3/2}}{\frac{1}{2}\Gamma(1/2)} \quad (\Gamma(1/2) = \sqrt{\pi})$$

$$= 4\pi^{3/2-\frac{1}{2}} = 4\pi$$

So,

$$I = \frac{2\pi^{w-2}}{\Gamma(w-2)} \int d^4\ell \underbrace{\int_0^\infty \frac{dL L^{2w-5}}{(L^2 + \ell^2 + \mu^2)}}_{\text{easy}} \quad \begin{array}{l} 2w-6 \geq 0 \\ w \geq 3 \end{array}$$

uv divergent for $w \geq 3$

IR divergence for $w \leq 1$ (check by $\mu = 0$).

Write $L^{2w-5} = \frac{1}{w-2} \frac{d}{dL^2} \left((L^2)^{w-2} \right)$

$$dL L^{2w-5} = L dL L^{2w-6} = \frac{1}{2} dL^2 (L^2)^{w-3}$$

Integrate by parts over L^2 . & discard the surface term.

$$I = \underbrace{\frac{\pi^{w-2}}{(w-2)\Gamma(w-2)}}_{\Gamma(w-1)} \int d^4\ell \int dL^2 (L^2)^{w-2} \left(-\frac{d}{dL^2}\right) \frac{1}{L^4 + \ell^2 + \mu^2}$$

Again, IR divergent for $w \leq 1$
uv " " $w \geq 1$

Do integral by parts once more —

$$I = \frac{\pi^{w-2}}{\Gamma(w)} \int d^4\ell \int dL^2 (L^2)^{w-1} \left(-\frac{d}{dL^2}\right)^2 \frac{1}{L^4 + \ell^2 + \mu^2}$$

well defined for $(0 < w < 1)$

We need to look for I' which extends the domain & can be analytically continued to $w = 2$.

$$\text{use } 1 = \frac{1}{(5)} \left(\frac{\partial L}{\partial L} + \frac{\partial \ell_\mu}{\partial \ell_\mu} \right) \quad \left\{ \frac{\partial \ell_\mu}{\partial \ell_\mu} = 4 \right\}$$

Integrate by parts —

$$\begin{aligned} I &= -\frac{2\pi^{w-2}}{5\Gamma(w)} \int d^4\ell \int dL^2 \left(\ell_\mu \frac{\partial}{\partial \ell_\mu} + 2L^2 \frac{\partial}{\partial L^2} + 1 \right) \cdot \left(\frac{(L^2)^{w-1}}{(L^2 + \ell^2 + \mu^2)^3} \right) \\ &= \frac{-3\mu^2}{w-1} \frac{2\pi^{w-2}}{\Gamma(w)} \int d^4\ell \int_0^\infty dL^2 \frac{(L^2)^{w-1}}{(L^2 + \ell^2 + \mu^2)^4} \end{aligned}$$

Now, the only divergence is from the pole at $w = 2$.

Check,

$$\begin{aligned} I &= \int \frac{d^{2w}\ell}{(\ell^2 + \mu^2)^6} \longrightarrow I \\ I &= \frac{\pi^{w-2}}{\Gamma(w-1)} \int d^4\ell \int_0^\infty dL^2 (L^2)^{w-2} \left(-\frac{d}{dL}\right) \left(\frac{1}{L^4 + \ell^2 + \mu^2} \right) \\ I(w=2) &= \int d^4\ell \frac{(-1)}{(L^2 + \mu^2 + \mu^2)^6} \Big|_0^\infty = \int d^4\ell \frac{1}{(\ell^2 + \mu^2)^6} \end{aligned}$$

use,

$$\int \frac{d^{2w}\ell}{\ell^2 + \mu^2} = \frac{\pi^w \Gamma(1-w)}{\Gamma(1)} \cdot \frac{1}{(\mu^2)^{1-w}}$$

← expand around $w = 2$. (use $\epsilon = 2 - w$)

$$\Gamma(-n + \epsilon) = \frac{(-1)^n}{n!} \left(\frac{1}{\epsilon} + \psi(n+1) + \frac{\epsilon}{2} \left(\frac{\pi^2}{3} + \psi^2(n+1) - \psi'(n+1) \right) + O(\epsilon^2) \right)$$

$$\frac{d\psi(s)}{ds} = \frac{d}{ds} \ln \Gamma(s)$$

$$\psi(n+1) = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \gamma$$

↘ euler maclaurian const.

$$\psi(1) = -\gamma = -0.577$$

$$\psi'(n+1) = \frac{\pi^2}{6} - \sum_1^n \frac{1}{k^2} \quad \psi'(1) = \pi^2/6$$

So,

$$\lim_{w \rightarrow 2} \int \frac{d^{2w} \ell}{\ell^2 + \mu^2} = -\pi^2 \mu^2 \left(\frac{1}{2-w} + \psi(2) + O(\epsilon) \right)$$

↖ Blows up.

↘ quadratically divergent for $w = 2$

10 Lecture 10–11

(The scan of these notes is a single notebook marked “10, 11”. The break between the two lectures falls at “2-loop diagrams” below.)

ϕ^4 theory renormalisation —

$$S = \int d^{2w} x \left(\frac{1}{2} (\partial \phi)^2 - \frac{m^2 \phi^2}{2} - \frac{\lambda}{4!} \phi^4 (\mu^2)^{2-w} \right)$$

Here ‘ m ’ is bare mass in place of ‘ μ ’.

μ is introduced to make action dimensionless. & λ dimensionless in each ‘ $2w$ ’ dim.

(margin working)

$$\int d^{2w} x \left(-m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 (\mu^2)^n \right) \quad [\varphi] = \frac{2w-2}{2} = w-1$$

$$1) \quad 2 + 2[\varphi] - 2w = 0$$

$$2) \quad -2w + 4[\varphi] + n(2) = 0$$

$$-2w + 4(w-1) + 2n = 0$$

$$2n = 2w - 4(w-1)$$

$$2n = -2w + 4$$

$$\underline{n = 2 - w}$$

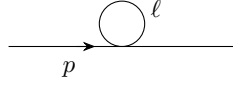
Minor changes in Feynman Rules :

- 1) Scalar product of k_1 & k_2 is summed over $2w$ components.

$$k_1 \cdot k_2 = k_1^0 k_2^0 - \left(\vec{k}_1 \cdot \vec{k}_2 \right)_{\text{in } (2w-1) \text{ dim}}$$

2) vertex $-\lambda \longrightarrow -\lambda (\mu^2)^{2-w}$

let's look at tadpole diagram —



$$T = \frac{1}{2} (-\lambda) (\mu^2)^{2-w} \int \frac{d^{2w} \ell}{(2\pi)^w} \frac{1}{(\ell^2 + m^2)} \quad \text{---(1)}$$

$\frac{1}{2}$: from where ?

(Digression) Integrals over arbitrary dimensions —

$$I_N = \int d^N \ell F(\ell)$$

goto polar coordinates — $(\ell_1, \ell_2 \dots \ell_N) \downarrow (L, \phi, \theta_1, \theta_2, \dots \theta_{N-2})$

$$L^2 = \ell_\mu \ell_\mu \quad (\text{Euclidean space})$$

$$d^N \ell = L^{N-1} dL (d\phi) (\sin \theta_1 d\theta_1) (\sin^2 \theta_2 d\theta_2) \dots (\sin^{N-2} \theta_{N-2} d\theta_{N-2})$$

$$0 \leq L < \infty, \quad 0 \leq \phi < 2\pi, \quad 0 \leq \theta_i < \pi$$

Standard Formulae —

$$\int_0^{\pi/2} (\sin t)^{2x-1} (\cos t)^{2y-1} dt = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)} \quad \begin{array}{l} \text{Re } x > 0 \\ \text{Re } y > 0 \end{array}$$

$\{A(2) \text{ of Ramond QFT}\}$

For $y = \frac{1}{2}$

$$\int_0^\pi \sin^k t dt = 2 \int_0^{\pi/2} \sin^k t dt = \frac{\sqrt{\pi} \Gamma(\frac{k+1}{2})}{\Gamma(\frac{k+2}{2})}$$

So,

$$I_N = \frac{\pi^{N/2}}{\Gamma(N/2)} \int_0^\infty dx x^{\frac{N-2}{2}} F(x) \quad (x = L^2) \quad \text{---(2)}$$

$$\left\{ \begin{array}{l} \text{comparing ① \& ②} \quad \frac{d^{2w} \ell}{(2\pi)^w} = dx \\ \frac{1}{\ell^2 + m^2} = x^{\frac{N-2}{2}} F(x) \end{array} \right.$$

For $F = (x+a)^{-A}$; $A = 2, 3, \dots$

$$\int_0^\infty dx \frac{x^{\frac{N-2}{2}}}{(x+a^2)^A} = (a^2)^{-A+\frac{N}{2}} \int_0^\infty dy y^{\frac{N-2}{2}} (1+y)^{-A}$$

$$\text{Recall, } B\left(\frac{N}{2}, A - \frac{N}{2}\right) = \frac{\Gamma(N/2) \Gamma(A - N/2)}{\Gamma(A)} = \int_0^\infty dy y^{\frac{N}{2}-1} (1+y)^{-A}$$

$$\begin{cases} \operatorname{Re}(N/2) > 0 \\ \operatorname{Re}(A - N/2) > 0 \end{cases}$$

$$\int \frac{d^N \ell}{(\ell^2 + a^2)^A} = \frac{\pi^{N/2} \Gamma(A - N/2)}{\Gamma(A)} \frac{1}{(a^2)^{A - N/2}}$$

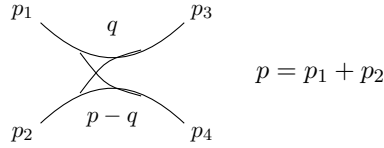
We take this formula valid to complex values of N by analytic continuation.

Another integral we need — (we are doing in euclidean signature)

$$\int d^N \ell \frac{\ell_\mu \ell_\nu}{(\ell^2 + 2p \cdot \ell + a^2)^A} = \frac{\pi^{N/2}}{\Gamma(A) (a^2 - p^2)^{A - \frac{N}{2}}} \left[\Gamma\left(A - \frac{N}{2}\right) p_\mu p_\nu + \frac{\delta_{\mu\nu}}{2} \Gamma\left(A - 1 - \frac{N}{2}\right) (a^2 - p^2) \right]$$

↓ Arbitrary vector.

Such Formula arrives in



$p = p_1 + p_2$

Considering A can take complex values.

$$\begin{aligned} T &= \frac{1}{2} (-\lambda) (\mu^2)^{2-w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \\ &= \frac{-\lambda m^2}{2 (4\pi)^2} \left(\frac{4\pi^2 \mu^2}{m^2} \right)^{2-w} \Gamma(1-w) \end{aligned}$$

expand it ① ↘ pole at $w = 1$.

(margin working)

$$\begin{aligned} \Gamma(m) &= (m-1)! = \int_0^\infty t^{m-1} e^{-t} dt \\ \Gamma(1-w) &= (1-w-1)! = \int_0^\infty t^{-w} e^{-t} dt \\ &= \int_0^\infty t^{-w} \left(1 - t + \frac{t^2}{2!} - \frac{t^3}{3!} + \dots \right) dt \\ &= \int_0^\infty dt \left(t^{-w} - t^{1-w} + \frac{t^{2-w}}{2!} + \dots \right) = \left(\frac{t^{1-w}}{-w+1} - \frac{t^{1-w}}{2-w} + \frac{t^{2-w}}{(3-w)2!} + \dots \right) \Big|_0^\infty \end{aligned}$$

Expand about $2-w$ ($\equiv \epsilon$)

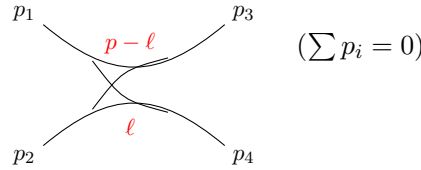
$$T = \frac{-\lambda m^2}{32\pi^2} \left[1 + (2-w) \ln \left(\frac{4\pi\mu^2}{m^2} \right) + \dots \right] \left[\frac{-1}{2-w} - \psi(2) + \dots \right]$$

pole term from lowest series $\downarrow$

$$= \frac{\lambda m^2}{32\pi^2} \left(\frac{1}{2-w} + \psi(2) - \underbrace{\ln \left(\frac{m^2}{4\pi^2 \mu^2} \right)}_{\text{finite part}} + O(2-w) \right)$$

- The divergence in $T \sim$ simple pole (in ϵ)
- The finite part is completely arbitrary. (depends on μ^2) as different people have diff. values of μ .

Next look at $\Gamma^{(4)}$



$$= \frac{+1}{2} (-\lambda)^2 (\mu^2)^{4-2w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \frac{1}{(\ell - p)^2 + m^2}$$

$$(P = p_1 + p_2)$$

Feynman parameterization —

$$\frac{1}{D_1^{a_1} D_2^{a_2} \cdots D_K^{a_K}} = \frac{\Gamma(a_1 + a_2 + \cdots + a_K)}{\Gamma(a_1) \Gamma(a_2) \cdots \Gamma(a_K)} \int_0^1 \cdots \int_0^1 dx_1 dx_2 \cdots dx_K$$

$$\times \frac{\delta(1 - (x_1 + x_2 + \cdots + x_K))}{(D_1 x_1 + D_2 x_2 + \cdots + D_K x_K)^{(a_1 + a_2 + \cdots + a_K)}} \left(x_1^{a_1-1} x_2^{a_2-1} \cdots x_K^{a_K-1} \right)$$

$$\frac{1}{(\ell^2 + m^2) ((\ell + p)^2 + m^2)} = \int_0^1 dx \frac{1}{[\ell^2 + m^2 + 2\ell \cdot p(1-x) + p^2(1-x)]^2}$$

$$\downarrow (\ell - p)^2 \quad \textcircled{?}$$

$$D_r = (\ell')^2 + m^2 + p^2 x(1-x);$$

$$d^{2w} \ell' = d^{2w} \ell$$

So,  is $= \frac{\lambda^2}{2} (\mu^2)^{4-2w} \int_0^1 dx \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2 + p^2 x(1-x))^2}$

use,

$$\int \frac{d^{2w} \ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + M^2 + 2\ell \cdot p)^A} = \frac{\Gamma(A-w)}{(4\pi)^w \Gamma(A)} \frac{1}{(M^2 - p^2)^{A-w}}$$

$$(\text{bubble}) = \frac{\lambda^2}{2} (\mu^2)^{4-2w} \int_0^1 dx \frac{\Gamma(2-w)}{(4\pi)^w} \frac{1}{(M^2 + p^2 x(1-x))^{2-w}}$$

Expand in powers of ϵ ($= 2 - w$) :

$$(\text{bubble}) = (\mu^2)^{2-w} \frac{\lambda}{32\pi^2} \left(\frac{1}{2-w} + \psi(1) - \int_0^1 dx \ln \left(\frac{M^2 + p^2 x(1-x)}{4\pi\mu^2} \right) \right)$$

The finite part depends upon μ^2 & p (ext. momentum)

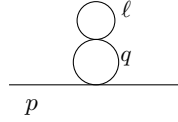
We know

$$\int_0^1 dx \ln \left(1 + \frac{4x}{a}(1-x) \right) = -2 + \sqrt{1+a} \ln \left(\frac{\sqrt{1+a}+1}{\sqrt{1+a}-1} \right)$$

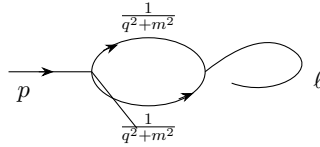
Using above,

$$\begin{aligned} (\text{bubble}) &= (\mu^2)^{2-w} \left(\frac{\lambda}{32\pi^2} \right) \left[\frac{1}{2-w} + \psi(1) + 2 + \ln \frac{4\pi\mu^2}{m^2} \right. \\ &\quad \left. - \left(1 + \frac{4m^2}{p^2} \right)^{1/2} \ln \left(\frac{\sqrt{1 + \frac{4m^2}{p^2}} + 1}{\sqrt{1 + \frac{4m^2}{p^2}} - 1} \right) \right] (\ln 4\pi m) \end{aligned}$$

2-loop diagrams — To be done!



$$= \frac{\lambda^2}{4} (\mu^2)^{4-2w} \int \frac{d^{2w}\ell}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \int \frac{d^{2w}q}{(q^2 + m^2)(q^2 + m^2)}$$



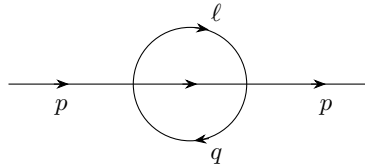
Subleading part $\downarrow$

$$= \frac{\lambda^2 m^2}{1024 \pi^4} \left(\frac{1}{(2-w)^2} + \frac{1}{(2-w)} \left(2 \ln \frac{4\pi\mu^2}{m^2} + \psi(2) + \psi(1) \right) \right)$$

$\swarrow$ arbitrary

$$\begin{aligned} &+ 2 \ln^2 \frac{4\pi\mu^2}{m^2} + 2 \ln \frac{4\pi\mu^2}{m^2} (\psi(2) + \psi(1)) \\ &+ \frac{1}{2} \left\{ (\psi(2) + \psi(1))^2 + \frac{2\pi^2}{3} - \psi'(2) - \psi'(1) + O(2-w) \right\} \end{aligned}$$

The other 2-loop diagram —



$$\Sigma(p) = \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \left\{ \frac{d^{2w} \hat{\ell}}{(2\pi)^{2w}} \frac{d^{2w} \check{q}}{(2\pi)^{2w}} \frac{1}{(\ell^2 + m^2)} \frac{1}{(q^2 + m^2)} \frac{1}{((P - \ell + q)^2 + m^2)} \right\}$$

$$\text{use} \quad \frac{1}{4w} \left(\frac{\partial \ell_\mu}{\partial \ell_\mu} + \frac{\partial q_\mu}{\partial q_\mu} \right) = 1 \quad \begin{pmatrix} \downarrow & 2w \\ \downarrow & 2w \end{pmatrix}$$

(margin working)

$$\begin{aligned} \Sigma(p) &= \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \left\{ \frac{d^{2w} \hat{\ell}}{(2\pi)^{2w}} \frac{d^{2w} \check{q}}{(2\pi)^{2w}} \cdot \left(\ell_\mu \frac{\partial}{\partial \ell_\mu} + q_\mu \frac{\partial}{\partial q_\mu} \right) \right. \\ &\quad \times \frac{1}{(\ell^2 + m^2)} \frac{1}{(q^2 + m^2)} \frac{1}{((p + q - \ell)^2 + m^2)} \end{aligned}$$

Integrate by parts

$$\begin{aligned} \Sigma(p) &= \frac{1}{4w} \frac{\lambda^2}{6} (\mu^2)^{4-2w} \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \\ &\quad \times \frac{3m^2 + p \cdot (p + q - \ell)}{(q^2 + m^2) (\ell^2 + m^2) [(q + p - \ell)^2 + m^2]^2} \\ &= \frac{1}{2w-3} \frac{\lambda^2}{6} (\mu^2)^{4-2w} [3m^2 K(p) + p^\mu K_\mu(p)], \text{ where} \\ K(p) &= \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \frac{1}{(q^2 + m^2) (\ell^2 + m^2) ((q + p - \ell)^2 + m^2)} \\ K_\mu(p) &= \int \frac{d^{2w} \ell}{(2\pi)^{2w}} \int \frac{d^{2w} q}{(2\pi)^{2w}} \frac{(p + q - \ell)_\mu}{(q^2 + m^2) (\ell^2 + m^2) ((q + \ell - p)^2 + m^2)} \end{aligned}$$

Using Feynman parameterization —

$$K(p) = \int d^{2w} \tilde{\ell} d^{2w} \tilde{q} \frac{1}{(q^2 + m^2)} \int_0^1 dx \frac{1}{\left(\underbrace{\ell^2 + m^2 + (p + q)^2 x(1-x)}_{M^2} \right)^2}$$

do the ℓ -integral first

$$\int d^{2w} \ell \frac{1}{(\ell^2 + M^2 + 2\ell \cdot p)^A} = \frac{\Gamma(A-w)}{(4\pi)^w \Gamma(A)} \frac{1}{(M^2 - p^2)^{A-w}}$$

Comparing $A=2$ ($\Gamma(2) = 1! = 1$)

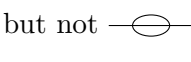
$$K(p) = \frac{\Gamma(2-w)}{(4\pi)^w} \int_0^1 dx \int d^{2w} \check{q} \frac{1}{(q^2 + m^2)^2} \frac{1}{[m^2 + (p + q)^2 x(1-x)]^{2-w}}$$

use Feynman parameterization once more to deal with

$$\frac{1}{(q^2 + m^2)^2} \frac{1}{((p + q)^2 x(1-x) + m^2)^{2-w}}$$

$$K(p) = \frac{\Gamma(2-w)}{(4\pi)^w} \int_0^1 dx (x(1-x))^{w-2} \int_0^1 dy y^{1-w}(1-y) \cdot \int \frac{d^{2w}q}{(2\pi)^w} (q^2 + p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right))^{w-4}$$

- DR allows us to do more than one loop calculations.

- In φ^3 theory counterterm kills  but not  & .

do the q integral first,

$$K(p) = \frac{\Gamma(4-2w)}{(4\pi)^{2w}} \int_0^1 dx (x(1-x))^{w-2} \int_0^1 dy y^{1-w}(1-y) \times \left(p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right)^{-2\epsilon}$$

Expand around $\epsilon = 0$ ($w = 2$).

We have to isolate poles from finite part.

$$\begin{aligned} K(p) &= \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \int_0^1 dx (x(1-x))^{-\epsilon} \int_0^1 dy y^{-1+\epsilon}(1-y) \\ &\times \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right]^{-2\epsilon} \\ &= \text{use } y^{-1+\epsilon} = \frac{1}{\epsilon} \frac{d}{dy} y^\epsilon, \text{ integrate by parts} \\ K(p) &= \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \frac{1}{\epsilon} \int dx (x(1-x))^{1-\epsilon} \int_0^1 dy y^\epsilon \times \left\{ 1 + 2\epsilon(1-y) \frac{d}{dy} \right. \\ &\times \ln \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right] \\ &\times \left. p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right\}^{-2\epsilon} \end{aligned}$$

Expand the integral about $\epsilon = 0$

$$K(p) = \frac{\Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \frac{1}{\epsilon} (1 + \epsilon - 2\epsilon \ln m^2 + O(\epsilon^2))$$

Now, Evaluate $K_\mu(p)$ similarly

{Mathew's & walker metn}

Since we are working in Euclidean metric. ($K_\mu = K^\mu$).

$$\begin{aligned} p^\mu K_\mu(p) &= \frac{p^2 \Gamma(2\epsilon)}{(4\pi)^{4-2w}} \int_0^1 dx (x(1-x))^{1-\epsilon} \int_0^1 dy y^\epsilon (1-y) \\ &\times \left[p^2 y(1-y) + m^2 \left(1 - y + \frac{y}{x(1-x)}\right) \right]^{-2\epsilon} \end{aligned}$$

Again expand around $\epsilon = 0$.

$$p^\mu K_\mu(p) = \frac{p^2 \Gamma(2\epsilon)}{(4\pi)^{4-2\epsilon}} \left(\frac{1}{2} + O(\epsilon) \right)$$

Putting All together :

$$\Sigma(p) = \frac{-\lambda^2}{6(16\pi^2)^2} \left[\frac{3m^2}{2\epsilon^2} + \frac{3m^2}{\epsilon} \left(\frac{1}{2} + \psi(1) + \ln \frac{4\pi^2 \mu^2}{m^2} \right) + \frac{p^2}{4\epsilon} + \text{finite term} \right]$$

Renormalisation —

poles in ‘ ϵ ’ can be eliminated order by order in λ .

$$\left(\hat{\lambda} \equiv \frac{\lambda}{16\pi^2} \right), \quad \hat{m}^2 = \frac{m^2}{16\pi^2 \mu^2}$$

Look at the tadpole digram —

$$\begin{array}{c} \text{---} \bigcirc \text{---} \\ \\ = m^2 \frac{\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + \psi(2) - \ln \hat{m}^2 + O(\epsilon) \right) \quad \text{---} \textcircled{3} \end{array}$$

$\nwarrow$ divergent as $\epsilon \rightarrow 0$

finite $F_1(\epsilon, m^2)$

Original lagrangian

$$\mathcal{L} = \frac{1}{2} (\partial\varphi)^2 - \frac{m^2 \varphi^2}{2} - \frac{\lambda}{4} \varphi^4 - \underbrace{g \varphi^2}_{\mathcal{L}_{int}}$$

$$\begin{array}{c} \text{bare mass} \\ \longleftarrow \quad \frac{\times}{g} \end{array}$$

(Choose coff g such that Counter term kills off divergence in $\text{---} \bigcirc \text{---}$ digram.)

Actually, Bare $\mathcal{L}$ itself is not defined as $\epsilon \rightarrow 0$ bare parameters depend on ϵ .

Add an extra term

$$\frac{m^2}{4} \hat{\lambda} \left(\frac{1}{\epsilon} + F_1(\epsilon, m^2) \right) \varphi^2$$

, where F_1 is arbitrary (dimensionless) function, analytic as $\epsilon \rightarrow 0$.

Puzzle :

Finite part is dependent on renormalization scheme, but we expect it to be same, wheather we do cut off renormalisation or DR. So how to reconcile ?

Our $\mathcal{L}$ looks like :

$$\mathcal{L} = \frac{1}{2} (\partial\varphi)^2 - \frac{m^2 \varphi^2}{2} - \frac{\lambda}{4!} \varphi^4 + \underbrace{\frac{m^2 \hat{\lambda}}{4} \left(\frac{1}{\epsilon} + F_1 \right) \varphi^2}_{\text{Tree level contribution}}$$

$\downarrow$ Tree level contribution

This $\mathcal{L}$ has extra Feynman rule —

$$\text{---}\times\text{---} = -\frac{1}{2} m^2 \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) \quad \text{---(4)}$$

Lets compute $\Gamma^{(2)}$ or inverse propagator —

$$\Gamma_{\text{new}}^{(2)} = \text{---} + \text{---}\bigcirc\text{---} + \text{---}\times\text{---} + O(\lambda)$$

eq ③ ↓

$$\begin{aligned} &= p^2 + m^2 + \frac{m^2 \hat{\lambda}}{2} \left[\frac{1}{\epsilon} + \psi(2) - \ln \hat{m}^2 + O(\epsilon) \right] \\ &\quad - \frac{1}{2} m^2 \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) + O(\hat{\lambda}) \\ &= p^2 + m^2 + \frac{m^2 \hat{\lambda}}{2} (\psi(2) - \ln \hat{m}^2 - F_1) + O(\hat{\lambda}) \\ &= p^2 + m^2 \left(1 + \frac{\hat{\lambda}}{2} (\psi(2) - \ln \hat{m}^2 - F_1) \right) + O(\hat{\lambda}) \end{aligned}$$

→ means upto one loop.

2 pt. Fun. is now finite (upto order $\hat{\lambda}$)

The extra term is called counterterm $\left(\frac{m^2}{4} \hat{\lambda} \left(\frac{1}{\epsilon} + F_1 \right) \right)$

look at $O(\lambda^2)$

The 1 P I 4 pt. function —

$$\begin{array}{c} \text{---}\bigcirc\text{---} \\ \downarrow \\ \text{Not 1PI } \textcolor{red}{?} \end{array} = \text{---}\times\text{---} + \text{---}\times\text{---} + \text{---}\times\text{---} + \text{---}\times\text{---} + O(\lambda^3)$$

$$\begin{aligned} \Gamma^{(4)}(p_1, p_2, p_3, p_4) &= -\mu^{2\epsilon} \lambda \left(1 - \frac{3\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + \psi(1) + 2 + \ln \hat{m}^2 \right. \right. \\ &\quad \left. \left. - \frac{1}{3} A(s, t, u) + O(\epsilon) \right) \right) + O(\lambda^3) \end{aligned}$$

$$A(s, t, u) = \sum_{z=s, t, u} \left(1 + \frac{4m^2}{z} \right)^{1/2} \ln \left(\frac{\left(1 + \frac{4m^2}{z} \right)^{1/2} + 1}{\left(1 + \frac{4m^2}{z} \right)^{1/2} - 1} \right)$$

s, t, u are mendelstam variables

$$\begin{aligned}s &= (p_1 + p_2)^2 = (p_1 + p_2) \cdot (p_1 + p_2) \\ t &= (p_1 + p_3)^2 \\ u &= (p_1 + p_4)^2\end{aligned}$$

$$\left(\Gamma^{(4)} \text{ diverges as } \frac{1}{\epsilon} \right)$$

So we add a counterterm to $\mathcal{L}$ of the form —

$$\text{Counterterm} = \frac{1}{4} \mu^{2\epsilon} \lambda \frac{3\hat{\lambda}}{2} \left(\frac{1}{\epsilon} + G_1(\epsilon, m^2) \right) \varphi^4$$

↓ arbitrary but analytic as $\epsilon \rightarrow 0$.

One more Feynman rule gets to be added due to counterterm.

$$\begin{aligned}& \text{Diagram: four lines meeting at a central point} = -\frac{3}{2} \mu^{2\epsilon} \lambda \hat{\lambda} \left(\frac{1}{\epsilon} + G_1 \right) \\ \Gamma_{\text{(new)}}^4 &= \text{Diagram: four lines meeting at a central point} + \text{Diagram: two loops on two lines} + \text{Diagram: two lines with a bubble} + \text{Diagram: two lines with a bubble and a cross} + \text{Diagram: four lines meeting at a central point}\end{aligned}$$

$$\begin{aligned}\Gamma_{\text{new}}^{(4)}(p_1, p_2, p_3, p_4) &= -\mu^{2\epsilon} \lambda \left(1 - \frac{3}{2} \hat{\lambda} (-G_1 + \psi(1) + 2 + \ln \hat{m}^2 \right. \\ &\quad \left. - \frac{1}{3} A(s, t, u) \right) + O(\epsilon) + O(\lambda^3).\end{aligned}$$

Now divergence is gone!

We need only to renormalize $\Gamma^{(2)}$ & $\Gamma^{(4)}$ then all 8 point & 16 point functions

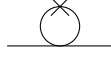
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Similarly we can study $\Gamma^{(2)}$ to 2 loops :

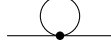
$$\begin{aligned}\text{Diagram: circle with diagonal lines} &= \text{Diagram: horizontal line} + \text{Diagram: circle on horizontal line} + \text{Diagram: cross on horizontal line} \\ &+ \text{Diagram: two circles on horizontal line} + \text{Diagram: large circle on horizontal line} + \text{Diagram: circle with cross on horizontal line} \\ &+ \text{Diagram: circle on horizontal line with a dot}\end{aligned}$$

counter term diagram

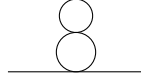
There are 2 new diagrams—



$$= \frac{m^2}{4} \hat{\lambda}^2 \left(\frac{1}{\epsilon^2} + \frac{1}{\epsilon} (\psi(1) + F_1 - \ln \hat{m}^2) + \dots \right)$$

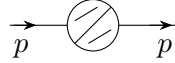


$$= \frac{3m^2}{4} \hat{\lambda}^2 \left(\frac{1}{\epsilon^2} + \frac{1}{\epsilon} (\psi(2) + G_1 - \ln \hat{m}^2) + \dots \right)$$



$$= -\frac{m^2}{4} \hat{\lambda}^2 \left(\frac{1}{\epsilon^2} + \frac{1}{\epsilon} (\psi(2) + \psi(1) - 2 \ln \hat{m}^2) + \dots \right)$$

Adding up we get—



$$= \frac{-\hat{\lambda}^2}{24\epsilon} p^2 + \frac{m^2 \lambda^2}{2} \left(\frac{1}{\epsilon^2} + \frac{1}{\epsilon} (F_1 + 3G_1 + 1) + \dots + O(\lambda^3) \right)$$

diverges as $\epsilon \rightarrow 0$

$$\ln \hat{m}^2 = \ln \frac{m^2}{4\pi\mu^2} \quad \text{has disappeared.} \quad \text{Check !}$$

The new mass counterterm feynman rule become,

$$-\frac{m^2}{2} \left(\frac{\hat{\lambda}^2}{\epsilon^2} + \frac{1}{\epsilon} \left(\hat{\lambda} + \frac{\hat{\lambda}^2}{4} (F_1 + 3G_1 + 1) + \hat{\lambda}^2 F_2 + \hat{\lambda} F_1 \right) \right)$$

$$\text{arbitrary } F^n \ \& \ \epsilon \ln \hat{m}^2 \quad \downarrow \quad (\rightarrow \text{finite as } \epsilon \rightarrow 0)$$

Counterterm in $\mathcal{L}$ is

$$\frac{m^2}{4} \varphi^2 \left(\frac{\hat{\lambda}^2}{\epsilon} + \frac{1}{\epsilon} \left(\hat{\lambda} + \frac{\hat{\lambda}^2}{4} (F_1 + 3G_1 + 1) + \hat{\lambda}^2 F_2 + \hat{\lambda} F_1 \right) \right)$$

The other infinity $\left(\frac{-\hat{\lambda}}{24\epsilon}p^2\right)$ is taken care by additional counterterm of form

$$\frac{1}{2}(\partial_\mu\varphi)(\partial^\mu\varphi)\left(\left(\frac{-\hat{\lambda}^2}{24\epsilon}\right)+\hat{\lambda}^2 H_2(\epsilon, m^2)\right)$$

↓

How to fix this pieces

does this piece has important physics information ?

We have eliminated UV divergence upto $O(\lambda^2)$ (i.e. upto 2 loops)

$$\mathcal{L}^{\text{ren}} = \mathcal{L}_{\text{Bare}} + \mathcal{L}_{ct} \quad \mathcal{L}_{ct} = \mathcal{L} \text{ (due to additional counterterms).}$$

$$\mathcal{L}^{\text{ren}} = \frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{m^2\varphi^2}{2} - \frac{\lambda}{4!}\mu^{2\epsilon}\varphi^4; \quad \{\epsilon = 2 - w\}$$

$$\mathcal{L}_{ct} = \frac{1}{2}A\partial_\mu\varphi\partial^\mu\varphi + \frac{1}{2}m^2B\varphi^2 + \frac{\lambda}{4!}C\mu^{2\epsilon}\varphi^4 \quad \text{---(3)}$$

$$\mathcal{L}_{\text{Bare}} = \frac{1}{2}\partial_\mu\varphi_0\partial^\mu\varphi_0 - \frac{m_0^2\varphi_0^2}{2} - \frac{\lambda_0}{4!}\mu^{2\epsilon}\varphi_0^4 \quad \text{---(4)}$$

$\mathcal{L}_{\text{ren}}$ is exactly like ' $\mathcal{L}$ ' (except in $\mathcal{L}$ we write $m \rightarrow m_{\text{bare}}, \lambda \rightarrow \lambda_{\text{bare}}$).

A, B, C are specially choosen so that the green's fun. are finite as $\epsilon \rightarrow 0$

$$\text{Notation : } \left. \begin{array}{l} A \longrightarrow Z_\varphi - 1 \\ B \longrightarrow Z_m \\ C \longrightarrow Z_\lambda \end{array} \right\} \text{ used in some books}$$

We redefine Bare fields & bare coupling to write $\mathcal{L}_{\text{ren}}$ in terms of bare fields & coupling

Write,

$$\mathcal{L}_{\text{ren}} = \frac{1}{2}\partial_\mu\varphi_0\partial^\mu\varphi_0 - \frac{1}{2}m_0^2\varphi_0^2 - \frac{\lambda_0}{4!}\varphi_0^4 \quad \longleftarrow \text{eq}^n (3) + (4)$$

where,

$$\begin{aligned} \varphi_0 &= (1+A)^{1/2}\varphi \equiv Z_\varphi^{1/2}\varphi \\ m_0 &= m^2\left(\frac{1-B}{1+A}\right) = m^2(1-B)Z_\varphi^{-1} \\ \lambda_0 &= \lambda\mu^{2\epsilon}\left(\frac{1-C}{(1+A)^2}\right) = \lambda\mu^{2\epsilon}(1-C)Z_\varphi^{-2} \end{aligned}$$

$\varphi_0, \lambda_0, m_0$ are bare field, bare coupling & bare mass which diverges as $\epsilon \rightarrow 0$, the renormalized ones do not.

Source J is also rescaled to $J_0 = Z_\varphi^{-1/2}J$,

$\mathcal{L}_{\text{ren}}$ gives us the Green's function, with m & λ replaced by m_0 & λ_0 .

$$\tilde{\Gamma}_0^{(n)}(p_1, \dots p_n; \lambda_0, m_0, \epsilon) = Z_\varphi^{-n/2} \tilde{\Gamma}^{(n)}(p_1, p_2, \dots p_n; m, \lambda, \mu, \epsilon)$$

↑ compute with bare quantities. (μ = mass scale to make λ dimensionless in each dimension.)

$\tilde{\Gamma}^{(n)}$ are finite as $\epsilon \rightarrow 0$.

There is no ' μ ' dependence on RHS. ! (as LHS does not.)

$$\mu \frac{\partial}{\partial \mu} \tilde{\Gamma}_0^{(n)}(p_1 p_2 \dots p_n; \lambda_0, m_0, \epsilon) = 0 = \mu \frac{\partial}{\partial \mu} \left(Z_\varphi^{-n/2} \tilde{\Gamma}^{(n)}(\{p_i\}; m, \lambda, \mu, \epsilon) \right)$$

$$\left(\mu \frac{\partial}{\partial \mu} + \mu \frac{\partial \lambda}{\partial \mu} \frac{\partial}{\partial \lambda} + \mu \frac{\partial m}{\partial \mu} \frac{\partial}{\partial m} - n \left(\frac{\mu}{2} \frac{\partial}{\partial \mu} \ln Z_\varphi \right) \right) \tilde{\Gamma}^{(n)} = 0$$

where the underbraced pieces are, in order, dimensionless; $\beta\left(\lambda, \frac{m}{\mu}, \epsilon\right); \frac{\mu}{2m} \frac{\partial m^2}{\partial \mu} \equiv \gamma_m;$
 $\gamma_d\left(\lambda, \frac{m}{\mu}, \epsilon\right) \uparrow$ field γ function.

So all objects shall be dimensionless.

These functions β, γ_m & γ_d are dimensionless and analytic as $\epsilon \rightarrow 0$. (they depend only on λ & $\frac{m}{\mu}$)

$\tilde{\Gamma}^{(n)}$ has n fields

$$\text{So mass dim. of } \tilde{\Gamma}^{(n)} = n \times \text{mass dim. of one field} = 4 - n - \epsilon(n - 2)$$

(margin working)

$$= w - 2 + 1; \quad w - 1 = 1 - \epsilon; \quad \epsilon = 2 - w$$

$$G^{(n)} = \langle \phi(x) \dots \phi(x_n) \rangle$$

$$S = \int d^{2w} \left(-\frac{m^2 \varphi^2}{2} - \frac{\lambda}{4!} \mu^x \varphi^4 \right)$$

$$1) 0 = -2w + 2 + 2[\varphi] \Rightarrow 2[\varphi] = 2w - 2; = w - 1$$

$$2) 0 = [\lambda]^0 + x + 4[\varphi] \quad (\epsilon = 2 - w)$$

$$x = -4(w - 1) = -4w + 4 = -4(2 - \epsilon) + 4 = -4 - 4\epsilon !$$

$$\left[\Gamma^{(n)} \right] = n \cdot (w - 1) = n((2 - \epsilon) - 1)$$

$$G^{(n)} = V(x_1 x_2 \dots x_n) G^{(2)} G^{(2)} \dots G^{(2)}$$

Rescaling the momenta

$$\tilde{\Gamma}^{(n)}(\{s p_i\}; m, \lambda, \mu, \epsilon)$$

rescaling p_i — we find

$$\left(\mu \frac{\partial}{\partial \mu} + s \frac{\partial}{\partial s} + m \frac{\partial}{\partial m} \right) \tilde{\Gamma}^{(n)} = (4 - n + \epsilon(n - 2)) \times \tilde{\Gamma}^{(n)}(\{s p_i\}; m, \lambda, \mu, \epsilon)$$

with $\mu \frac{\partial}{\partial \mu} \longrightarrow \frac{\partial}{\partial(\ln \mu)}$

Eliminate $\mu \frac{\partial}{\partial \mu}$ & get,

$$\begin{aligned} \left(-s \frac{\partial}{\partial s} + \beta \left(\lambda, \frac{m}{\mu} \right) \frac{\partial}{\partial \lambda} + \left(\gamma_m \left(\lambda, \frac{m}{\mu} \right) - 1 \right) m \frac{\partial}{\partial m} - n \gamma_d \left(\lambda, \frac{m}{\mu} \right) + (4 - n) \right) \\ \times \tilde{\Gamma}^{(n)}(\{s p_i\}; m, \lambda, \mu) = 0 \end{aligned}$$

Renormalization Group eqⁿ or Gell mann – Law eqⁿ.

————— Mid sem —————

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The idea is to express bare parameters as a laurent series in the renormalized parameters

$$\begin{aligned} \lambda_0 &= \mu^{2\epsilon} \left(a_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) + \sum_{k=1}^{\infty} \frac{a_k(\lambda, m/\mu)}{\epsilon^k} \right) \\ m_0^2 &= m^2 \left(b_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) + \sum_{k=1}^{\infty} \frac{b_k(\lambda, m/\mu)}{\epsilon^k} \right) \\ Z_\varphi &= C_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) + \sum_{k=1}^{\infty} \frac{C_k(\lambda, m/\mu)}{\epsilon^k} \end{aligned}$$

Where,

$$a_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) = \lambda \left(1 + \frac{3}{2} \hat{\lambda} G_1 \right) + O(\lambda^2)$$

and

$$b_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) = 1 + \frac{1}{2} \left(\hat{\lambda} F_1 + \hat{\lambda}^2 F_2 + \hat{\lambda}^2 H_2 \right) + O(\lambda^3)$$

and

$$C_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) = 1 - \hat{\lambda}^2 H_2(m/\mu, \epsilon) + O(\lambda^3)$$

$$a_1(\lambda, m/\mu) = \frac{3}{2} \frac{\lambda^2}{16\pi^2} + O(\lambda^3)$$

$$b_1(\lambda, m/\mu) = \frac{1}{2} \left(\hat{\lambda} + \frac{\hat{\lambda}^2}{4} (F_1 + 3G_1 + 1) + \frac{\hat{\lambda}^2}{24} \right) + O(\lambda^5)$$

$$b_2(\lambda, m/\mu) = \frac{1}{2} \hat{\lambda}^2 + O(\lambda^3)$$

$$C_1(\lambda, m/\mu) = -\frac{\hat{\lambda}^2}{24} + O(\lambda^3)$$

$$\text{-----} \times \text{-----}$$

ϕ^6 theory in 4d, do as before: renormalize $m_0^2, \lambda_4, Z_\phi, \lambda_6$ (the interactions being $\frac{\lambda}{4!}\phi^4$ and $\frac{\lambda}{6!}\phi^6$).

Now look at 8 pt. function. (it turns out this is not finite ! at loop level (not at tree level).)

$$V \sim \frac{m^2 \phi^2}{2!} + \lambda_4 \frac{\phi^4}{4!} + \lambda_6 \frac{\phi^6}{6!}$$

↓
automatically generates ϕ^4 interaction, at tree level.

$$\text{Diagram: two external lines crossing with a loop} \implies 2 \rightarrow 2 \text{ scattering.}$$

we need $\frac{\lambda_8}{8!}\phi^8$ when we add counterterm to get rid of infinities in ϕ^6 theory. Just like we needed $\frac{\lambda}{4!}\phi^4$ in bare lagrangian to get rid of this diagram.

But in ϕ^4 theory 4pt, 6pt, 8pt... fun. are all finite once we have renormalized m_0, λ_4, Z_ϕ .

So For ϕ^6 theory we need infinite no. of coupling to be added in $\mathcal{L}$ to renormalize 2pt, 4pt, 6pt, 8pt, 10pt ... even pt. functions

$$V = \sum \lambda_{2n} \phi^{2n} \quad \text{to be added in } \mathcal{L}_0.$$

- Renormalizable theory requires to make finite couplings renormalized.
- Non renormalized theory requires renormalization of infinite couplings. (we need infinite measurements). **Not a good theory (has weak predictive power).** For low momenta it is good at providing

Remember we wrote down —

$$\tilde{\Gamma}_0^{(n)}(p_1, p_2 \dots p_n; \lambda_0, m_0, \epsilon) = Z_\phi^{-n/2} \tilde{\Gamma}^{(n)}(p_1, p_2 \dots p_n; m, \lambda, \mu, \epsilon)$$

↓ Bare n -pt fun.

↓ renormalized n -pt fun.

Now, vary $\tilde{\Gamma}^{(n)}$ w.r.t. physical mass ' m ', this gives us Callen – Symanzik eqⁿ.

$$\text{-----} \times \text{-----}$$

Summary —

We were looking at renor. of φ^4 theory (to 2 loops). We introduced arbitrary scale μ .

- We arrived at analytic fun. in counterterms ($F_1, F_2 \dots G_1, H_2 \dots$) These fun. are the finite part of the counterterms.

$$\mathcal{L}_{\text{ren}} = \mathcal{L} + \mathcal{L}_{ct}.$$

The finite part of $\mathcal{L}_{ct}$ can be fixed by defining the parameters appearing in $\mathcal{L}$.

Scheme A :

physical mass = zero of 2 pt fun (2pt fun. with zero momenta)
bare mass = coeff of φ^2 .

$$\begin{aligned} (\eta = (+1, 1, 1, 1) \text{ check it !}) \\ \tilde{\Gamma}^{(2)}(p, m_A) &= p^2 + m_A^2 \quad (\text{mean } \vec{p}^2 = 0) \\ \tilde{\Gamma}^{(4)}(p_1 p_2 p_3 p_4) &= -\mu^{2\epsilon} \lambda_A \quad \text{at } p_i = 0. \end{aligned}$$

As long as, $m_A^2 \neq 0$ (non-zero mass theory) above prescription is well defined.

This fixes F_1, G_1, H_2 for

$$\begin{aligned} \tilde{\Gamma}^{(2)}(p) &= p^2 + m^2 \left(1 + \frac{\hat{\lambda}}{2} (\psi(2) - \ln \hat{m}^2 - F_1) \right) + O(\lambda^2) \\ F_1^{(A)} &= \psi(2) - \ln \hat{m}_A^2 \end{aligned}$$

Similarly,

$$G_1^A = \psi(1) - \ln \hat{m}_A^2 ; \quad H_2^A = 0$$

Scheme B :

One can change the subtraction pt, as long as it does not mess up the analytic continuation to minkowski space, or other IR divergences.

As long as we perform this subtraction on Euclidean Green function (vertex fun), it will give us a space like subtraction in Minkowski space.

$$(p^0, 0) \longrightarrow (0, \vec{p})$$

time like space like

$$\begin{aligned}
\tilde{\Gamma}^{(2)}(p, m_B) &= p^2 + m_B^2 \quad \text{at } p^2 = M^2 \\
\tilde{\Gamma}^{(4)}(\{p_j\}) &= -\lambda \mu^{2\epsilon} \quad \text{at } p_i p_j = M^2 \left(\delta_{ij} - \frac{1}{4} \right) \\
&\quad (s = t = u = M^2) \\
F_1^B &= \psi(2) - \ln \hat{m}_B^2, \quad H_2^B = 0 \\
G_1^B &= \psi(1) - \ln \hat{m}_B^2 - \int_0^1 dx \ln \left(1 + \frac{M^2}{m_B^2} x(1-x) \right)
\end{aligned}$$

In scheme B, the renormalisation eqⁿ are not easy to solve, except deep in euclidean region where masses can be neglected.

Scheme C :

We set all the finite parts of counterterm to be zero (order by order in λ).

$$0 = F_1^C = F_2^C = G_1^C = F_2^C$$

The advantage of scheme ‘C’ becomes clear when we look at $a_0, b_0, C_0 \dots$

$$\begin{aligned}
a_0 \left(\lambda, \frac{m}{\mu}, \epsilon \right) &= \lambda \left(1 + \frac{3}{2} \hat{\lambda} G_1 \right) + O(\lambda^2), \\
b_0() &= ()
\end{aligned}$$

All the coefficients are independent of m .

This prescription is called mass – independent renormalization

$$\lambda_0 = \mu^{2\epsilon} \left(\lambda + \sum_{k=1}^{\infty} \frac{a_k(\lambda)}{\epsilon^k} \right) \quad \text{etc.}$$

differentiate w.r.t $\mu \frac{\partial}{\partial \mu}$ at fixed λ_0 we get,

$$0 = 2\epsilon \left(\lambda + \sum_{k=1}^{\infty} \frac{a_k(\lambda)}{\epsilon^k} \right) + \mu \frac{\partial \lambda}{\partial \mu} \left(1 + \sum_{k=1}^{\infty} \frac{a'_k(\lambda)}{\epsilon^k} \right)$$

here λ & $\frac{\partial \lambda}{\partial \mu}$ are analytic at $\epsilon = 0$. so we get—

$$\begin{aligned}
\mu \frac{\partial \lambda}{\partial \mu} &= -2\epsilon \lambda - 2a_1(\lambda) + 2\lambda a'_1(\lambda) \\
\lim_{\epsilon \rightarrow 0} \mu \frac{\partial \lambda}{\partial \mu} &= \beta(\lambda) = -2 \left(1 - \lambda \frac{\partial}{\partial \lambda} \right) a_1(\lambda) \\
\left(1 - \lambda \frac{\partial}{\partial \lambda} \right) a_{k+1}(\lambda) &= a'_k(\lambda) \left(1 - \lambda \frac{\partial}{\partial \lambda} \right) a_1(\lambda)
\end{aligned}$$

$$a_1(\lambda) = \frac{3}{2} \frac{\lambda^2}{16\pi^2} + O(\lambda^2)$$

$$\Rightarrow \quad \mu \frac{\partial \lambda}{\partial \mu} = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3)$$

integrate, we get

$$\int \frac{\partial \lambda}{\frac{3\lambda^2}{16\pi^2} + O(\lambda^3)} \simeq \int \frac{\partial \mu}{\mu}$$

$$\lambda(\mu) = \lambda_s \frac{1}{1 - \frac{3}{16} \lambda_s \ln \frac{\mu}{\mu_s}} ; \quad \lambda_s \equiv \lambda(\mu_s)$$

13 A Short Break from Renormalisation

Quantised EM field interacting with

(Intro to infrared problem in QED)
 $\hookrightarrow$ topic of current research.

Warmup : SHO with a forcing function (1-d) — To build necessary technology

Recall —

1) Time evolution in QM is given by

$$|\psi(t)\rangle = T(t, t_0) |\psi(t_0)\rangle \quad \text{---(1)}$$

$$TT^\dagger = T^\dagger T = I \quad (\text{Probability conservation})$$

$$T(t, t_0) = T(t, t_1) T(t_1, t_0)$$

$$T(t, t_0)^{-1} = T(t_0, t)$$

Also eqⁿ (1) requires

$$|\psi(t)\rangle|_{t=t_0} = |\psi(t_0)\rangle \quad \text{for } t = t_0$$

$$\text{i.e.} \quad T(t_0, t_0) = I$$

$$T(t + \epsilon, t) = 1 - \frac{i}{\hbar} \epsilon H(t) \quad \text{--- defⁿ of Hamiltonian}$$

$H(t)$ is defined as leading operator for time evolution operator (when expanded as first order in ϵ). ($\epsilon \downarrow$ time step)

$$i\hbar \frac{d}{dt} T(t, t_0) = H(t) T(t, t_0)$$

for H to be time independent —

$$T(t, t_0) = e^{-iH(t-t_0)/\hbar}$$

Pictures in QM —

1. Schrödinger picture — observables (operators) are fixed & wavefun / state evolve
2. Heisenberg picture — wave fun are fixed but observables evolve with time
3. Interacting picture — (Dirac picture) — Both states & observables evolve via convenient operators T_0 or H_0 .

Heisenberg picture —

(2) define $U(t) = T(0, t)$

This operator doesn't change state —

$$\begin{aligned} \text{i.e. } U(t) |\psi(t)\rangle &= U(t) T(t, 0) |\psi(0)\rangle \\ &= T(0, t) T(t, 0) |\psi(0)\rangle \\ &= I |\psi(0)\rangle \end{aligned}$$

define

$$\begin{aligned} \bar{L}(t) &= U(t) L U^\dagger(t) \\ i\hbar \frac{d}{dt} \bar{L} &= [\bar{L}, \bar{H}] + i\hbar \frac{\partial \bar{L}}{\partial t} \end{aligned}$$

(3) For dirac picture —

Choose $U(t)$ a solⁿ of $i\hbar \frac{\partial U}{\partial t} = -U H_0$

$$\begin{aligned} \bar{L} &= U(t) L U^\dagger(t) \\ i\hbar \frac{d}{dt} \bar{L} &= [\bar{L}, \bar{H}_0] + i\hbar \frac{\partial \bar{L}}{\partial t} \end{aligned}$$

where as states change

$$i\hbar \frac{d}{dt} |\bar{\psi}(t)\rangle = U(t) (H - H_0) U^\dagger(t) |\bar{\psi}(t)\rangle$$

where,

$$\begin{aligned} |\bar{\psi}(t)\rangle &= U(t) |\psi(t)\rangle \\ H - H_0 &= V = \text{interaction} \\ i\hbar \frac{d}{dt} |\bar{\psi}(t)\rangle &= \bar{V}(t) |\bar{\psi}(t)\rangle \quad \left(\bar{V} = U V U^\dagger \right) \end{aligned}$$

_____ × _____

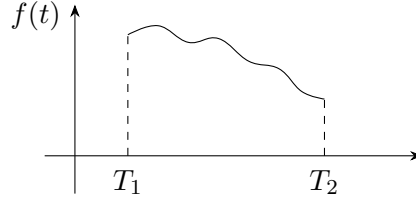
For

$$\begin{aligned} H &= \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 - q Q(t) - p P(t) \\ H &= \hbar \omega \left(a^\dagger a + \frac{1}{2} \right) + f(t) a + f^*(t) a^\dagger \end{aligned}$$

$$f(t) = -\sqrt{\frac{\hbar}{2m\omega}} Q(t) + i\sqrt{\frac{\hbar m\omega}{2}} P(t)$$

We are interested in

$$\begin{aligned} f &\neq 0 & T_1 < t < T_2 \\ f &= 0 & t < T_1 \quad \& \quad t > T_2 \end{aligned}$$



$$H = \hbar\omega \left(\bar{a}^\dagger(t) \bar{a}(t) + \frac{1}{2} \right) + f(t) \bar{a}(t) + f^*(t) \bar{a}^\dagger(t)$$

$$i\hbar \frac{d\bar{a}}{dt} = [\bar{a}, \bar{H}] = [\bar{a}, H] \quad (\bar{H} = H) \quad \left\{ \text{in Heisenberg picture } \hat{H}_H = \hat{H}_S \right\}$$

$$i\hbar \frac{d\bar{a}}{dt} = \hbar\omega \bar{a} + f^*(t) \quad \text{---(2)}$$

$$(t_0 = 0)$$

$$\bar{a}(t) = \bar{a} e^{-\frac{i\omega}{\hbar}t} - \frac{i}{\hbar} \int_0^t e^{-i\omega(t-t')} f^*(t') dt'$$

we can solve eqⁿ (2) using greens fun.

$$\left(\frac{d}{dt} + i\omega \right) G(t, t') = \delta(t - t')$$

$$\bar{a}(t) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} G(t, t') f^*(t') dt'$$

Retarded green's fun. —

$$G_R(t - t') = \theta(t - t') e^{-\frac{i\omega}{\hbar}(t-t')}$$

$$\theta(x) = 0 ; x < 0, \quad = 1 ; x > 0$$

Advanced greens fun —

$$G_A(t - t') = -\theta(t' - t) e^{-i\omega(t-t')} = -G_R(t - t')$$

So,

$$\bar{a}(t) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} G_R(t - t') f^*(t') dt' \quad \text{vanishes for } t < T_1$$

$$= -\frac{i}{\hbar} \int_{-\infty}^{\infty} G_A f^* \quad \text{vanishes if } t > T_2$$

let $\bar{a}_{\text{in}}$ be solⁿ of $\frac{d\bar{a}}{dt} + i\omega\bar{a} = 0$ which coincide with solⁿ using G_R (G_A) resp.

$$\bar{a}(t) = \bar{a}_{\text{in}}(t) - \frac{i}{\hbar} \int_{-\infty}^{\infty} G_R(t-t') f^*(t') dt' \quad (\bar{a}_{\text{in}} \rightarrow \bar{a} e^{-i\omega t})$$

$$\Rightarrow \quad \bar{a}_{\text{out}}(t) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} G_A(t-t') f^*(t') dt'$$

$$\bar{a}_{\text{out}}(t) = \bar{a}_{\text{in}}(t) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} (G_R - G_A) f^*(t') dt'$$

$$\bar{a}_{\text{out}}(t) = \bar{a}_{\text{in}}(t) - \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' e^{-i\omega(t-t')} f^*(t')$$

$$(\bar{a}_{\text{out}} e^{-i\omega t}, \quad \bar{a}_{\text{in}} e^{-i\omega t})$$

$$\bar{a}_{\text{out}} = \bar{a}_{\text{in}} - \frac{i}{\hbar} \int_{-\infty}^{\infty} dt' e^{i\omega t'} f^*(t')$$

$$\overbrace{\hspace{10em}} \text{just a complex number } \tilde{g}(\omega).$$

Compute $|\langle n_{\text{out}} | 0_{\text{in}} \rangle|^2$

Interaction Picture —

Take $H_0 = \hbar\omega (a^\dagger a + \frac{1}{2})$ & $V = f(t)a + f^*(t)a^\dagger$

Then,

$$\bar{V}(t) = e^{i\hbar\omega(a^\dagger a + \frac{1}{2})t} V e^{-i\hbar\omega(a^\dagger a + \frac{1}{2})t}$$

use BCH formula to find

$$\begin{aligned} & \begin{cases} e^{i\omega\hbar a^\dagger a} a e^{-i\omega\hbar a^\dagger a} \\ e^{-i\omega\hbar a^\dagger a} a^\dagger e^{-i\omega\hbar a^\dagger a} \end{cases} \\ &= f(t) a e^{-i\omega t} + f^* a^\dagger e^{i\omega t} \end{aligned}$$

Time evolution $\bar{T} = U T U^\dagger$

where $\bar{T}$ obeys :

$$i\hbar \frac{d}{dt} \bar{T}(t) = \bar{V}(t) \bar{T}(t)$$

Since $\bar{V} = \bar{V}(t)$ time dependent, we have $\bar{T}$ as time ordered exponential —

$$\begin{aligned} \bar{T}(t, t_1) &= \mathcal{T} \left\{ \exp \left(-\frac{i}{\hbar} \int_{t_1}^t \bar{V}(t') dt' \right) \right\} \quad (\mathcal{T} : \text{Time ordering}) \\ &= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{t_1}^t \cdots \int_{t_1}^t dt'_1 \cdots dt'_n \mathcal{T}(\bar{V}(t'_1) \bar{V}(t'_2) \cdots \bar{V}(t'_n)) \end{aligned}$$

For one $V(t)$ which is linear in a & $a^\dagger$ we can exactly solve for $\bar{T}(t, t_1)$

$$\bar{T}(t, t_1) = \lim_{N \rightarrow \infty} e^{V_N} e^{V_{N-1}} \cdots e^{V_2} e^{V_1}$$

$$\left(\begin{array}{ccc} \cdots & \downarrow & \downarrow \\ & T(t_1+2\epsilon, t_1+\epsilon) & T(t_1+\epsilon, t_1) \end{array} \right)$$

Where

$$V_k = -\frac{i}{\hbar} \int_{t_1+(k-1)\epsilon}^{t_1+k\epsilon} \bar{V}(t') dt' \quad (t - t_1 = N\epsilon)$$

We have

$$\bar{T}(t, t_1) = \dots e^{V_k} e^{V_{k-1}} \dots$$

using BCH formula—

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]} \quad \text{if} \quad [A, [A, B]] = [B, [A, B]] = 0$$

We combine $e^{V_k} e^{V_{k-1}}$ terms using above

$$\begin{aligned} [\bar{V}(t'), \bar{V}(t'')] &= f(t') f^*(t'') e^{-i\omega(t'-t'')} - f^*(t') f(t'') e^{i\omega(t'-t'')} \\ \bar{T}(t, t_1) &= \exp \left(-\frac{i}{\hbar} \int_{t_1}^t \bar{V}(t') dt' - \frac{1}{2\hbar^2} \int_{t_1}^t \int_{t_1}^t dt' dt'' [\bar{V}(t'), \bar{V}(t'')] \right) \\ &= e^{i\beta(t, t_1)} e^{-\zeta^*(t, t_1) \bar{a} + \zeta(t, t_1) \bar{a}^\dagger} \\ \zeta(t, t_1) &= -\frac{i}{\hbar} \int_{t_1}^t e^{i\omega t'} f^*(t') dt' \\ g(\omega) &= -i\hbar \zeta^*(T_1, T_2) = -i\hbar \zeta^*(-\infty, \infty) \end{aligned}$$

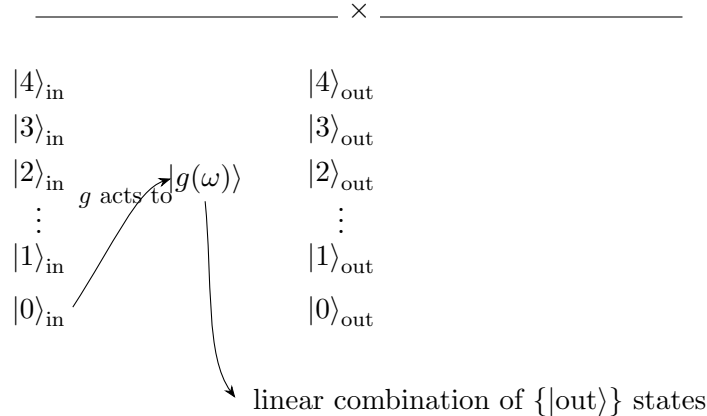
Defining scattering operator $S = \bar{T}(-\infty, \infty)$

Then

$$\begin{aligned} S &= e^{i\beta} \exp \left(-\frac{i}{\hbar} g(\omega) a - \frac{i}{\hbar} g^* a^\dagger \right) \\ |\langle n | S | 0 \rangle|^2 &= \text{Transition probability for } |0\rangle \longrightarrow |n\rangle \\ &= \frac{1}{n!} \left| \frac{g(\omega)}{\hbar} \right|^{2n} e^{-|g(\omega)|^2/\hbar^2} \\ &= \frac{(\langle n \rangle)^n}{n!} e^{-\langle n \rangle} \quad (\text{poisson distribution}) \end{aligned}$$

If we don't turn off $f(t)$ fast enough the fourier transform $g(\omega)$ does not exist !

$\hookrightarrow$ IR divergence !



So $|0\rangle_{\text{in}} \neq |0\rangle_{\text{out}}$

But physically speaking $|0\rangle_{\text{in}}$ seems same !